The Blow-Up Formula in Prismatic Cohomology: Nygaard Compatibility, Iterated Towers, Geometric Applications, and Cycle Map

Version 14 — Final Audit + Strong Prismatic Cycle

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May 2026

Abstract. Let X be a regular scheme, proper and smooth over O_K, and let $i : Z \hookrightarrow X$ be a regular closed immersion of codimension c. We prove a canonical isomorphism

$$R\Gamma_{\mathrm{Prism}}(\mathrm{Bl}_Z X) \;\simeq\; R\Gamma_{\mathrm{Prism}}(X) \;\oplus\; \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r]$$

in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$, φ-equivariant and compatible with the Nygaard filtration $\mathcal{N}^\bullet$.

Scope of results by part.

- **Part I (Sections 1–7): Core.** Theorem 1.1, Nygaard compatibility (Proposition 5.1), integral descent (Proposition 6.3). All results are complete as stated.
- **Part II (Sections 8–9): Iterated Blow-Ups.** The single-step formula extends to any finite regular blow-up tower by induction. No new hypotheses are introduced.
- **Part III (Sections 10–11): Geometric Applications.** Applied to minimal resolutions of A_n and D_n surface singularities. These are geometric corollaries of Part II.
- **Part IV (Sections 12–16): Prismatic Cycle Map.** A cycle class map $\mathrm{cl}_r^{\mathrm{Prism}} : \mathrm{CH}^r(X) \to H^{2r}_{\mathrm{Prism}}(X)(r)$ is constructed and studied. All sections are complete as stated. Section 13 proves the integral local-global obstruction theorem in full: Lemma 13.1 establishes $H^i_{\mathrm{Prism}}(X)[p^\infty] = 0$ under (B) alone; Lemma 13.2 establishes freeness of $H^i_{\mathrm{Prism}}(X)$ as an A_{inf}-module under (A)+(B).
- **Appendix A: Compatibility with the Lefschetz trace formula.** The zeta-function factorization $Z(\widetilde{X}_0/\mathbb{F}_q, t) = Z(X_0/\mathbb{F}_q, t) \cdot \prod_{r=1}^{c-1} Z(Z_0/\mathbb{F}_q, q^r t)$ follows from Proposition 6.3 via the crystalline comparison.
- **Part V (Sections 17–20): Strong Prismatic Cycle.** *New in Version 14.* The prismatic Hodge locus and strongly filtered classes are introduced (Section 17). Conjecture 18.1 (prismatic Hodge conjecture) is stated and proved for $r = 1$ for abelian schemes and K3 surfaces (Section 18); the case $r \geq 2$ is open. Torsion-injectivity of $\mathrm{cl}_r^{\mathrm{Prism}}$ is proved for $r = 1$ unconditionally under (A)+(B); for $r \geq 2$ a conditional result is given under an additional explicit hypothesis (C) (Section 19). Stability of the Hodge locus under blow-up and revised outlook (Section 20).

Three-tier status of results.

Tier	**Content**
Closed and complete	Parts I–IV; cycle map; blow-up compatibility; crystalline/de Rham comparison; zeta factorization.
Proved in base cases	Conjecture 18.1 for $r = 1$ (abelian schemes, K3 surfaces); torsion-injectivity of $\mathrm{cl}_1^{\mathrm{Prism}}$.
Open / conditional	Conjecture 18.1 for $r \geq 2$; torsion-injectivity for $r \geq 2$ (conditional on hypothesis (C)); prismatic motivic realization functor.

The integral statement of Part I requires two propagatable hypotheses: a smooth $W_2(k)$-lift with local Frobenius lifts (A), and p-torsion-free Hodge cohomology (B).

Part I
The Blow-Up Formula: Core Results

Scope. *The results of this part are complete as stated. Every statement is either unconditional or requires only hypotheses (A)+(B).*

Sections 1–7 contain the closed nucleus: Theorem 1.1, Nygaard compatibility, and integral descent.

1 Introduction

1.1 Setup and main result

Let p be a prime, K a complete p-adic field with ring of integers $\mathcal{O}_K$ and perfect residue field k. Write $W(k)$ for the Witt vectors of k and $A_{\mathrm{inf}} = W(\mathcal{O}_C^\flat)$ for Fontaine's period ring, where C is a completed algebraic closure of K.

We use $R\Gamma_{\mathrm{Prism}}(X)$ for the prismatic complex of a p-adic formal scheme X as in [2, 3]. We write $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$ for the derived category of perfect A_{inf}-complexes equipped with a semilinear Frobenius φ and a descending Nygaard filtration $\mathcal{N}^\bullet$.

Theorem 1.1 (Main Theorem)**.** *Let X be regular, proper, and smooth over $\mathcal{O}_K$, and let $i\colon Z \hookrightarrow X$ be a regular closed immersion of codimension $c \geq 1$. Let $\pi\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ be the blow-up with exceptional divisor $E \simeq \mathbb{P}(\mathcal{N}_{Z/X})$.*

1. (Rational, unconditional.) *There is a canonical isomorphism in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}[1/p])$:*
$$R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \;\simeq\; R\Gamma_{\mathrm{Prism}}(X) \;\oplus\; \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r], \tag{1}$$
φ-equivariant and compatible with $\mathcal{N}^\bullet$.

2. (Integral, under (A)+(B).) *Under the hypotheses of Proposition 6.3, the isomorphism holds in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$.*

Remark. *Part (1) is established in Sections 3–4, together with Nygaard compatibility (Proposition 5.1). Part (2) is Proposition 6.3. Sections 3–5 are entirely rational.*

Remark. *Hypothesis (B) ensures the graded pieces*
$$\mathrm{gr}^j_{\mathcal{N}} H^i_{\mathrm{Prism}}(X) \otimes_{A_{\mathrm{inf}},\theta} W(k) \simeq H^{i-j}(X, \Omega^j_{X/\mathcal{O}_K})$$
are torsion-free, hence free $W(k)$-modules. The classical blow-up formula then gives integral isomorphisms on each graded piece. Hypothesis (A) ensures degeneration of the Hodge spectral sequence at E_1 over $W_2(k)$ [5], assembling graded isomorphisms into a full filtered isomorphism. The two hypotheses are asymmetric in their effect: without (B), the Hodge groups may have p-torsion on one side of the comparison but not the other, breaking condition (iii) of Lemma 6.1 at every Hodge degree simultaneously. Without (A), only condition (ii) of Lemma 6.1 may fail, a more localised damage. Neither can be dropped; see Warning .

1.2 Overview of the manuscript

Part II (Sections 8–9). Theorem 8.2 extends (1) to any finite regular blow-up tower $X = X_0 \leftarrow \cdots \leftarrow X_n$. Hypotheses (A)+(B) propagate at every step via Lemma 6.2. Corollary 9.2 handles pairwise disjoint centers.

Part III (Sections 10–11). Theorem 8.2 is applied to minimal resolutions of A_n and D_n surface singularities (Propositions 10.2 and 10.3). The prismatic cohomology of the resolution equals $R\Gamma_{\text{Prism}}(S) \oplus A_{\text{inf}}(-1)[-2]^{\oplus n}$ in both cases; the distinction between types is carried by the intersection pairing on exceptional classes, not by the direct-sum decomposition.

Part IV (Sections 12–16). A cycle class map $\mathrm{cl}_r^{\text{Prism}}\colon \mathrm{CH}^r(X) \to H^{2r}_{\text{Prism}}(X)(r)$ is constructed from the prismatic fundamental class of [3, Sec. 9]. Sections 12 and 14–15 are complete. Section 13 identifies a local-global obstruction in $\mathrm{Ext}^1_{A_{\text{inf}}}(H^{2r-1}_{\text{Prism}}(X)(r), A_{\text{inf}})$; the integral statement is conditional (see the scope note at the head of Part IV).

Appendix A. The blow-up formula implies the factorization $Z(\widetilde{X}_0/\mathbb{F}_q, t) = Z(X_0/\mathbb{F}_q, t) \cdot \prod_{r=1}^{c-1} Z(Z_0/\mathbb{F}_q, q^r t)$, recovered from the prismatic side via the crystalline comparison.

1.3 Precise inputs from [2, 3]

- [2, Thm. 1.8]: Perfectness of $R\Gamma_{\text{Prism}}(X)$, finiteness of p-torsion, comparisons with A_{cris} and B^+_{dR}.
- [2, Prop. 4.13]: $H^i_{\text{Prism}}(X)[p^\infty]$ is finite with explicit exponents.
- [2, Thm. 1.8(4)]: Hodge–Tate comparison $\mathrm{gr}^j_{\mathcal{N}} H^i_{\text{Prism}}(X) \otimes_{A_{\text{inf}},\theta} W(k) \simeq H^{i-j}(X, \Omega^j_{X/\mathcal{O}_K})$ (functorial in X).
- [3, Sec. 7]: Prismatic Chern classes $c_r^{\text{Prism}}(E)$, their φ-equivariance, and the fact that $c_1^{\text{Prism}}(L) \in \mathcal{N}^1 H^2_{\text{Prism}}(X)(1)$.
- [3, Sec. 9]: Purity/Gysin isomorphism $R\Gamma_{\text{Prism},Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$ for regular Z of codimension c, with φ- and $\mathcal{N}^\bullet$-compatibility.
- [2, Thm. 5.2]: Proper base change for $R\pi_*$.

2 Preliminaries

2.1 Prismatic cohomology: standing facts

For X proper smooth over $\mathcal{O}_K$, the complex $R\Gamma_{\text{Prism}}(X) \in D^{\varphi,N}_{\text{pris}}(A_{\text{inf}})$ satisfies:

(P1) **Perfectness.** Each $H^i_{\text{Prism}}(X)$ is finitely generated and p-adically complete over A_{inf}; $R\Gamma_{\text{Prism}}(X)$ is perfect [2, Thm. 1.8].

(P2) **Finite torsion.** There exists $N(X,i) \geq 0$ with $p^{N(X,i)} \cdot H^i_{\text{Prism}}(X)[p^\infty] = 0$ [2, Prop. 4.13].

(P3) **Crystalline comparison.** A canonical φ-equivariant isomorphism $R\Gamma_{\text{Prism}}(X) \otimes^L_{A_{\text{inf}}} A_{\text{cris}} \xrightarrow{\sim} R\Gamma_{\text{cris}}(X/W(k))$.

(P4) **de Rham comparison.** A canonical φ-equivariant isomorphism

$$R\Gamma_{\text{Prism}}(X) \otimes^L_{A_{\text{inf}}} B^+_{\text{dR}} \xrightarrow{\sim} R\Gamma_{\text{dR}}(X_K),$$

with $\mathcal{N}^\bullet$ corresponding to the Hodge filtration [2, Thm. 1.8].

(P5) **Hodge–Tate graded pieces.** For each $j \geq 0$:

$$\mathrm{gr}^j_{\mathcal{N}} H^i_{\text{Prism}}(X) \otimes_{A_{\text{inf}},\theta} W(k) \simeq H^{i-j}(X, \Omega^j_{X/\mathcal{O}_K}).$$

(P6) **Frobenius and Nygaard.** $\varphi(\mathcal{N}^s H^i_{\text{Prism}}(X)) \subseteq p^s H^i_{\text{Prism}}(X)$. The induced map

$$\varphi/p^s\colon \mathrm{gr}^s_{\mathcal{N}} H^i_{\text{Prism}}(X) \to H^i_{\text{Prism}}(X)$$

is injective after inverting p [3, Sec. 12].

(P7) **Faithful flatness.** $A_{\text{inf}}[1/p] \to A_{\text{cris}}[1/p]$ is faithfully flat [2, Lem. 3.23].

2.2 Cohomology with support

For $i\colon Z \hookrightarrow X$ a closed immersion with open complement $j\colon U := X \setminus Z$:

$$R\Gamma_{\mathrm{Prism},Z}(X) := \mathrm{Cone}\Big(R\Gamma_{\mathrm{Prism}}(X) \xrightarrow{j^*} R\Gamma_{\mathrm{Prism}}(U)\Big)\,[-1].$$

Remark (Filtered excision triangle)**.** *Since j^* lies in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$, the object $R\Gamma_{\mathrm{Prism},Z}(X)$ inherits φ and $\mathcal{N}^\bullet$ via:*

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism},Z}(X) := \mathrm{Cone}(\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \to \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(U))\,[-1].$$

This yields the filtered excision triangle

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism},Z}(X) \to \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \to \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(U) \xrightarrow{+1},$$

mapping termwise to the unfiltered triangle.

2.3 Prismatic Chern classes and the Tate twist convention

The prismatic first Chern class is constructed in [3, Sec. 7] and satisfies:

(C1) $\varphi(c_1^{\mathrm{Prism}}(L)) = p \cdot c_1^{\mathrm{Prism}}(L)$.

(C2) $c_1^{\mathrm{Prism}}(L) \in \mathcal{N}^1 H^2_{\mathrm{Prism}}(X)(1)$.

(C3) Under (P3), $c_1^{\mathrm{Prism}}(L) \mapsto c_1^{\mathrm{cris}}(L)$.

(C4) If $\alpha \in \mathcal{N}^s$, then $\alpha \cup c_1^{\mathrm{Prism}}(L) \in \mathcal{N}^{s+1}$.

Lemma 2.1 (Tate twist convention)**.** *For $M \in D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$ and $r \in \mathbb{Z}$: $\mathcal{N}^s(M(-r)) = \mathcal{N}^{s+r}(M)$.*

Proof. $x \in \mathcal{N}^s(M(-r))$ iff $\varphi_{M(-r)}(x) = p^{-r}\varphi_M(x) \in p^s M(-r)$, i.e. $\varphi_M(x) \in p^{s+r}M$, i.e. $x \in \mathcal{N}^{s+r}(M)$. $\square$

3 Two Key Lemmas

3.1 Projective bundle formula

Lemma 3.1 (Projective bundle formula)**.** *Let $q\colon P = \mathbb{P}(\mathcal{E}) \to Z$ be the projective bundle of a locally free sheaf of rank c, and set $\xi = c_1^{\mathrm{Prism}}(\mathcal{O}_P(1))$. The map*

$$\Phi\colon \bigoplus_{r=0}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r] \xrightarrow{\sim} R\Gamma_{\mathrm{Prism}}(P), \qquad (\alpha_r) \mapsto \sum_{r=0}^{c-1} q^*(\alpha_r) \cup \xi^r,$$

is an isomorphism in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$.

Proof. **(a) Rational isomorphism.** Apply $(-)\otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}}$. By (P3) and (C3), $\Phi\otimes A_{\mathrm{cris}}$ coincides with the classical projective bundle formula in crystalline cohomology [6, Exp. VII, Prop. 3.3]. Faithful flatness (P7) promotes this to $\Phi[1/p]$.

(b) Integral isomorphism. Source and target are perfect by (P1). For perfect complexes, f is an isomorphism iff $f[1/p]$ and all $H^i(f)[p^\infty]$ are. Part (a) handles $f[1/p]$. For the torsion, (P5) identifies $\mathrm{gr}^j_{\mathcal{N}} H^i_{\mathrm{Prism}}(P)$ with $H^{i-j}(P, \Omega^j)$, and the classical formula [6, Exp. VII] equates the p-torsion on both sides.

(c) φ-equivariance. On the source, φ acts on the r-th summand as $p^r \cdot \varphi_Z$. Since $\varphi(\xi^r) = (p\xi)^r = p^r\xi^r$ by (C1) and q^* commutes with φ: $\varphi(q^*(\alpha_r) \cup \xi^r) = p^r \cdot q^*(\varphi(\alpha_r)) \cup \xi^r$.

(d) $\mathcal{N}^\bullet$-compatibility. By Lemma 2.1: $\mathcal{N}^s(R\Gamma_{\mathrm{Prism}}(Z)(-r)) = \mathcal{N}^{s+r}(R\Gamma_{\mathrm{Prism}}(Z))$. Since $\xi \in \mathcal{N}^1$ by (C2), iterating (C4) gives $\xi^r \in \mathcal{N}^r$, and if $\alpha_r \in \mathcal{N}^{s-r}$ then $q^*(\alpha_r) \cup \xi^r \in \mathcal{N}^s$. The converse follows from (P5) and the classical formula on graded pieces. $\square$

3.2 Gysin isomorphism

Lemma 3.2 (Gysin isomorphism)**.** *Let $i\colon Z \hookrightarrow X$ be a regular closed immersion of codimension c. There is a canonical isomorphism in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$:*

$$R\Gamma_{\mathrm{Prism},Z}(X) \simeq R\Gamma_{\mathrm{Prism}}(Z)(-c)[-2c], \tag{2}$$

φ-equivariant and $\mathcal{N}^\bullet$-compatible.

Proof. **(a) Local Koszul model.** Locally, the ideal of Z is generated by a regular sequence $f_1, \dots, f_c \in \mathcal{O}_X$. Since $\mathcal{O}_{\mathrm{Prism}}$ is flat over $\mathcal{O}_X$ [3, Lem. 2.25], the sequence is regular in $\mathcal{O}_{\mathrm{Prism}}$, giving $R\Gamma_{\mathrm{Prism},Z\cap U}(U) \simeq R\Gamma_{\mathrm{Prism}}(Z \cap U)[-2c]$.

(b) Tate twist. The fundamental class $\eta_Z \in H^{2c}_{\mathrm{Prism},Z}(X)(c)$ of [3, Sec. 9] satisfies $\varphi(\eta_Z) = p^c\eta_Z$, giving the local isomorphism $R\Gamma_{\mathrm{Prism},Z\cap U}(U) \simeq R\Gamma_{\mathrm{Prism}}(Z \cap U)(-c)[-2c]$.

(c) Globalization. Compatibility on overlaps follows from Remark 2.1; the sheaf axiom gives the global isomorphism (2).

(d) φ-equivariance and $\mathcal{N}^\bullet$-compatibility. φ-equivariance is from [3, Sec. 9]. For the filtration: $\eta \in \mathcal{N}^s R\Gamma_{\mathrm{Prism},Z}(X)$ iff $\eta' \in \mathcal{N}^{s-c}R\Gamma_{\mathrm{Prism}}(Z)$, and by Lemma 2.1, $\mathcal{N}^{s-c}R\Gamma_{\mathrm{Prism}}(Z) = \mathcal{N}^s(R\Gamma_{\mathrm{Prism}}(Z)(-c))$. □

4 Proof of Theorem 1.1(1)

Proof of Theorem 1.1, part (1). All claims are in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}[1/p])$.

Step 1. π^* is split injective.

Classical vanishing. $\pi_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$ and $R^i\pi_*\mathcal{O}_{\widetilde{X}} = 0$ for $i > 0$ [6, Exp. VII, Prop. 3.3].

Transfer to prismatic cohomology. Let α_c denote the crystalline comparison (P3).

(i) $R\pi_*R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \otimes A_{\mathrm{cris}} \simeq R\pi_*R\Gamma_{\mathrm{cris}}(\widetilde{X}/W(k))$ [2, Cor. 5.3].

(ii) Classical vanishing gives $R\pi_*R\Gamma_{\mathrm{cris}}(\widetilde{X}/W(k)) \simeq R\Gamma_{\mathrm{cris}}(X/W(k))$ [4, Thm. 1.4].

(iii) Applying α_c^{-1} and (P7): $R\pi_*R\Gamma_{\mathrm{Prism}}(\widetilde{X})[1/p] \simeq R\Gamma_{\mathrm{Prism}}(X)[1/p]$.

So $(R\pi_*) \circ (\pi^*) = \mathrm{id}$ rationally, and π^* is split injective with complement $C := \mathrm{Cone}(\pi^*)[-1] \in D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}[1/p])$.

Step 2. Identification $C \simeq R\Gamma_{\mathrm{Prism},E}(\widetilde{X})$.

Consider the two filtered excision triangles (Remark 2.1):

$$R\Gamma_{\mathrm{Prism},E}(\widetilde{X}) \to R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \to R\Gamma_{\mathrm{Prism}}(X \setminus Z) \xrightarrow{+1},$$
$$R\Gamma_{\mathrm{Prism},Z}(X) \to R\Gamma_{\mathrm{Prism}}(X) \to R\Gamma_{\mathrm{Prism}}(X \setminus Z) \xrightarrow{+1},$$

where we used $\widetilde{X} \setminus E \simeq X \setminus Z$. Use the splitting $R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus C$ from Step 1 and form the diagram of long exact sequences. The rightmost column is an isomorphism, and the middle column is the identity on $H^i(R\Gamma_{\mathrm{Prism}}(X))$. The five-lemma gives $C \simeq R\Gamma_{\mathrm{Prism},E}(\widetilde{X})$.

Step 3. Computation of C.

Lemma 3.2 gives $R\Gamma_{\mathrm{Prism},E}(\widetilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(E)(-1)[-2]$. Lemma 3.1 applied to $q\colon E = \mathbb{P}(\mathcal{N}_{Z/X}) \to Z$ (rank c) gives

$$R\Gamma_{\mathrm{Prism}}(E) \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

Combining:

$$C \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r-1)[-2r-2] = \bigoplus_{r=1}^{c} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

Step 4. Elimination of the $r = c$ summand.

We claim the summand $R\Gamma_{\mathrm{Prism}}(Z)(-c)[-2c]$ belongs to the image of π^* and therefore does not contribute to C.

Let

$$\gamma\colon R\Gamma_{\mathrm{Prism}}(Z)(-c)[-2c] \xrightarrow{\sim} R\Gamma_{\mathrm{Prism},Z}(X) \to R\Gamma_{\mathrm{Prism}}(X)$$

be the Gysin map of Lemma 3.2. This is precisely the class of the strict transform of Z in X: it exists before the blow-up, as the image of the fundamental class $\eta_Z \in H^{2c}_{\mathrm{Prism},Z}(X)(c)$ under the natural map $R\Gamma_{\mathrm{Prism},Z}(X) \to R\Gamma_{\mathrm{Prism}}(X)$, and therefore lies in $R\Gamma_{\mathrm{Prism}}(X)$, not in the new cohomology C introduced by the blow-up. Concretely, applying $R\pi_*$ and the projection formula: $R\pi_*(q^*(\alpha)) = \alpha \cdot R\pi_*\mathcal{O}_E = \alpha \cdot i_*\mathcal{O}_Z$, which lies in $R\Gamma_{\mathrm{Prism}}(X)$, confirming that the $r = c$ summand is in the image of π^*. Discarding it:

$$C \simeq \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r],$$

completing the proof of (1). □

5 Nygaard Compatibility

Proposition 5.1 (Nygaard compatibility)**.** *The rational isomorphism* (1) *is compatible with* $\mathcal{N}^\bullet$*: for each* $s \geq 0$*,*

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} \mathcal{N}^{s+r} R\Gamma_{\mathrm{Prism}}(Z). \tag{3}$$

Lemma 5.2. *The maps* π^* *and* $R\pi_*$ *in* $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}[1/p])$ *are* $\mathcal{N}^\bullet$*-compatible.*

Proof. π^* commutes with φ, hence preserves $\mathcal{N}^\bullet$. $R\pi_*$ preserves $\mathcal{N}^\bullet$ by [2, Thm. 5.2]. □

Lemma 5.3 (Filtered splitting)**.** *The splitting* $R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus C$ *is* $\mathcal{N}^\bullet$*-compatible, giving* $\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \oplus \mathcal{N}^s C$*.*

Proof. Both π^* and $R\pi_*$ preserve $\mathcal{N}^\bullet$ by Lemma 5.2. □

Lemma 5.4 (Filtered identification of C)**.** *The identification* $C \simeq R\Gamma_{\mathrm{Prism},E}(\widetilde{X})$ *is* $\mathcal{N}^\bullet$*-compatible.*

Proof. Both excision triangles are $\mathcal{N}^\bullet$-compatible by Remark 2.1. Applying $\mathrm{gr}^s_{\mathcal{N}}(-)$ converts each triangle into a long exact sequence of Hodge cohomology groups via (P5). The five-lemma gives

$$\mathrm{gr}^s_{\mathcal{N}} C \xrightarrow{\sim} \mathrm{gr}^s_{\mathcal{N}} R\Gamma_{\mathrm{Prism},E}(\widetilde{X})$$

for all s; since both are perfect with bounded filtration, this implies a filtered isomorphism. □

Proof of Proposition 5.1. By Lemma 5.3:

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \oplus \mathcal{N}^s C.$$

By Lemma 5.4: $\mathcal{N}^s C \simeq \mathcal{N}^s R\Gamma_{\mathrm{Prism},E}(\widetilde{X})$. By Lemmas 3.2(d) and 2.1:

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism},E}(\widetilde{X}) \simeq \mathcal{N}^{s+1} R\Gamma_{\mathrm{Prism}}(E).$$

By Lemma 3.1(d) and re-indexing $r' = r + 1$:

$$\mathcal{N}^{s+1} R\Gamma_{\mathrm{Prism}}(E) = \bigoplus_{r=0}^{c-1} \mathcal{N}^{s+1+r} R\Gamma_{\mathrm{Prism}}(Z) = \bigoplus_{r'=1}^{c} \mathcal{N}^{s+r'} R\Gamma_{\mathrm{Prism}}(Z).$$

The $r' = c$ summand maps into $\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X)$ by the $\mathcal{N}^\bullet$-compatible projection formula argument (Lemma 5.2), leaving the sum over $r' = 1, \ldots, c-1$ as $\mathcal{N}^s C$, giving (3). □

6 Integral Descent

Lemma 6.1 (Integral criterion)**.** *Let $f\colon P \to Q$ be a map of perfect A_{inf}-complexes with Nygaard filtrations. Assume*:

(i) *$f[1/p]$ is an isomorphism;*

(ii) *the Nygaard spectral sequence $E_1^{j,i-j} = \mathrm{gr}^j_{\mathcal{N}} H^i(-) \otimes k \Rightarrow H^i(-) \otimes k$ degenerates at E_1 for both P and Q;*

(iii) *$\mathrm{gr}^j_{\mathcal{N}} H^i(f) \otimes k$ is an isomorphism for all i, j.*

Then f is an isomorphism in $D(A_{\mathrm{inf}})$.

Proof. Set $C = \mathrm{Cone}(f)$. By (i) and (P1), each $H^i(C)$ is killed by p^N for some N. Condition (ii) gives direct sum splittings $H^i(P) \otimes k \simeq \bigoplus_j \mathrm{gr}^j_{\mathcal{N}} H^i(P) \otimes k$ and similarly for Q, natural in f. By (iii), each summand map is an isomorphism, so $H^i(f) \otimes k$ is an isomorphism. The long exact sequence on cohomology mod p then gives $H^i(C) \otimes k = 0$, hence $H^i(C) = 0$. $\square$

Warning. *Degeneration of the Nygaard spectral sequence is not preserved under general exact triangles, so it is not assumed for the cone $C = \mathrm{Cone}(f)$. Condition (ii) is used only to split $H^i(P) \otimes k$ and $H^i(Q) \otimes k$; the conclusion $H^i(C) \otimes k = 0$ follows from the long exact sequence without any assumption on C.*

Lemma 6.2 (Blow-up inherits hypotheses)**.** *Assume* (A) *for X and* (B) *for X and Z. Then $\widetilde{X} = \mathrm{Bl}_Z X$ also satisfies* (A) *and* (B).

Proof. **Hypothesis (A) for $\widetilde{X}$.** Let $\mathcal{X}$ be a smooth $W_2(k)$-lift of X with locally compatible Frobenius lifts φ_B. The center Z lifts to a regularly embedded $\mathcal{Z} \hookrightarrow \mathcal{X}$ [7, Ch. 0, §10.2], the blow-up $\widetilde{\mathcal{X}}$ is smooth over $W_2(k)$ [6, Exp. VII, Prop. 3.3], and local Frobenius lifts are constructed chart by chart. On the chart $\widetilde{B}_1 = B[s_2, \ldots, s_c]/(t_2 - t_1 s_2, \ldots, t_c - t_1 s_c)$, writing $\varphi_B(t_i) = t_i^p + p\delta_i$, define:

$$\widetilde{\varphi}_1(b) := \varphi_B(b) \ (b \in B), \quad \widetilde{\varphi}_1(s_l) := s_l^p + p(\delta_l - \delta_1 s_l^p) t_1^{-p}, \quad l = 2, \ldots, c. \tag{4}$$

The term $p(\delta_l - \delta_1 s_l^p) t_1^{-p}$ is well-defined in $\widetilde{B}_1$: since $p^2 = 0$, it lies in $p\widetilde{B}_1[1/t_1] \cap \widetilde{B}_1 = p\widetilde{B}_1$. Verification that $\widetilde{\varphi}_1$ preserves the relation $t_l - t_1 s_l$, reduces to Frobenius mod p, and agrees on chart overlaps is a direct computation using $p^2 = 0$.

Hypothesis (B) for $\widetilde{X}$. The classical blow-up formula [6, Exp. VII, Prop. 3.3] gives

$$H^m(\widetilde{X}, \Omega^j) \simeq H^m(X, \Omega^j) \oplus \bigoplus_{r=1}^{c-1} H^{m-2r}(Z, \Omega^j). \tag{5}$$

Since both $H^m(X, \Omega^j)$ and $H^m(Z, \Omega^j)$ are p-torsion-free by (B), so is $H^m(\widetilde{X}, \Omega^j)$. $\square$

Remark (Consistency check: Lemma 6.2, Step 1)**.** *Two points warrant explicit confirmation. (a) Well-definedness of* (4)*. Since $p^2 = 0$, the product $p(\delta_l - \delta_1 s_l^p)$ lies in $p\widetilde{B}_1[1/t_1]$, and $p\widetilde{B}_1[1/t_1] \cap \widetilde{B}_1 = p\widetilde{B}_1$ (flatness of $\widetilde{B}_1$ over $W_2(k)$ and t_1 a non-zero-divisor mod p). So the term is a genuine element of $\widetilde{B}_1$. (b) Agreement on overlaps. On $\widetilde{B}_1[s_l^{-1}] = \widetilde{B}_l[s_1^{-1}]$, both $\widetilde{\varphi}_1$ and $\widetilde{\varphi}_l$ act as φ_B on B and both satisfy $\widetilde{\varphi}(s_j) \equiv s_j^p \pmod p$ with the same correction term (with t_1 and t_l exchanged). A direct substitution using $p^2 = 0$ confirms agreement.*

Proposition 6.3 (Rational $\Rightarrow$ integral)**.** *Assume* (A)+(B) *for both X and Z. Then the rational isomorphism* (1) *holds in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$.*

Proof. Let $f\colon P \to Q$ be the map of Theorem 1.1(1), with

$$P = R\Gamma_{\mathrm{Prism}}(\widetilde{X}), \qquad Q = R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

By Lemma 6.2, $\widetilde{X}$ satisfies (A)+(B).

Condition (i). $f[1/p]$ is an isomorphism by Theorem 1.1(1).

Condition (ii). By Lemma 6.2 and (A), Deligne–Illusie [5] applied to $\widetilde{X}$, X, and Z gives Hodge degeneration at E_1 over $W_2(k)$; hence the Nygaard spectral sequence degenerates at E_1 for P and Q by (P5).

Condition (iii). By (B) and Lemma 6.2, all Hodge groups are p-torsion-free. The Hodge–Tate comparison (P5) gives identifications

$$\mathrm{gr}^j_{\mathcal{N}} H^i(P) \otimes k \simeq H^{i-j}(\widetilde{X}, \Omega^j) \otimes k, \tag{6}$$

$$\mathrm{gr}^j_{\mathcal{N}} H^i(Q) \otimes k \simeq H^{i-j}(X, \Omega^j) \otimes k \oplus \bigoplus_{r=1}^{c-1} H^{i-j-2r}(Z, \Omega^j) \otimes k. \tag{7}$$

These identifications are natural in f: f is built from π^*, q^*, and cup product with ξ^r, each compatible with the Hodge–Tate comparison by [3, Sec. 4 and 7]. Therefore the diagram

$$\begin{array}{ccc} \mathrm{gr}^j_{\mathcal{N}} H^i(P) \otimes k & \xrightarrow{\mathrm{gr}^j_{\mathcal{N}} H^i(f) \otimes k} & \mathrm{gr}^j_{\mathcal{N}} H^i(Q) \otimes k \\ \downarrow (6)\simeq & & \downarrow (7)\simeq \\ H^{i-j}(\widetilde{X}, \Omega^j) \otimes k & \xrightarrow{\alpha^{i-j,j}} & H^{i-j}(X, \Omega^j) \otimes k \oplus \bigoplus_{r=1}^{c-1} H^{i-j-2r}(Z, \Omega^j) \otimes k \end{array}$$

commutes, where $\alpha^{i-j,j}$ is the map of the classical blow-up formula for Ω^j. By (5), $\alpha^{i-j,j}$ is an isomorphism, so condition (iii) holds.

By Lemma 6.1, f is an isomorphism. The φ- and $\mathcal{N}^\bullet$-compatibility are inherited from Theorem 1.1(1) and Proposition 5.1. $\square$

Remark (Consistency of the commutative diagram)**.** *The Hodge–Tate comparison of [3, Sec. 4] is the map of topoi $X_{\mathrm{Prism}} \to X_{\mathrm{dR}}$, natural in X. For any $\mathcal{O}_K$-morphism $g\colon Y \to X$, naturality gives commutativity of the g^*-squares on both sides, accounting for the π^* and q^* components of f. For the cup product $\cup\xi^r$: compatibility with the Hodge–Tate comparison follows from [3, Sec. 7], where c_1^{Prism} maps to c_1^{dR}. The bottom arrow $\alpha^{i-j,j}$ is therefore identified with the classical blow-up formula map, which is an isomorphism by (5).*

Warning (Where each hypothesis is used)**.** *In the proof above, hypothesis (A) enters for X and Z (to apply Deligne–Illusie to the summands of Q) and for $\widetilde{X}$ (supplied by Lemma 6.2). Hypothesis (B) guarantees torsion-free Hodge groups, making $\alpha^{i-j,j}$ an isomorphism of k-vector spaces. The two hypotheses are asymmetric in their damage when dropped: without (B), the p-torsion mismatch breaks condition (iii) at every Hodge degree simultaneously. Without (A), only condition (ii) — degeneration — may fail, a more localised effect.*

- *Without (A): condition (ii) of Lemma 6.1 may fail.*
- *Without (B): the Hodge groups may have p-torsion on one side but not the other, breaking condition (iii) at all Hodge degrees.*

Proposition 6.4 (Effective torsion bound)**.** *Without* (A) *or* (B)*: let t_j^X (resp. t_j^Z) be the smallest $n \geq 0$ with $p^n H^{i-j}(X, \Omega^j) = 0$ (resp. $H^{i-2r-j}(Z, \Omega^j_{\mathbb{P}}) = 0$); set $N_X(i) := \sum_j t_j^X + C_X$, $N_Z(m) := \sum_j t_j^Z + C_Z$ (constants from [2, Prop. 4.13]). Then*

$$p^N \cdot H^i_{\mathrm{Prism}}(\widetilde{X})[p^\infty] = 0, \qquad N \leq \max\Bigl(N_X(i), \max_{1 \leq r \leq c-1} N_Z(i-2r)\Bigr).$$

Under (A)+(B)*,* $N = 0$*. A minimal example where* $N > 0$ *occurs is a surface* X *with* $H^1(X, \Omega^1)$ *having* p*-torsion: in that case* $t_1^X \geq 1$ *and the bound gives* $N \geq 1$*, with equality when all other Hodge groups are* p*-torsion-free.*

7 Examples

Both examples below verify Nygaard compatibility and confirm hypotheses (A)+(B) hold. No claim goes beyond Theorem 1.1.

Example 7.1 (Blow-up of $\mathbb{P}^2$ at a point)**.** *$X = \mathbb{P}^2_{\mathcal{O}_K}$, $Z = \{x\}$ a rational point, $c = 2$. Formula* (1)*:*

$$R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(\mathbb{P}^2) \oplus A_{\mathrm{inf}}(-1)[-2].$$

By Lemma 2.1 and (C1): $\mathcal{N}^1(A_{\mathrm{inf}}(-1)) = A_{\mathrm{inf}}(-1)$, $\mathcal{N}^2(A_{\mathrm{inf}}(-1)) = 0$, $\varphi = p \cdot \mathrm{id}$. Hypotheses (A)+(B) hold; $N = 0$ and the formula is integral.

Example 7.2 (Kummer K3)**.** *$A = E_1 \times E_2$, $p > 2$, $K = \mathrm{Kum}(A)$ from 16 blow-ups with $c = 2$. Formula* (1)*:*

$$R\Gamma_{\mathrm{Prism}}(K) \simeq R\Gamma_{\mathrm{Prism}}((A/\mu)^{\mathrm{sm}}) \oplus \bigoplus_{16} A_{\mathrm{inf}}(-1)[-2].$$

Each summand lies in $\mathcal{N}^1 H^2$, consistent with exceptional classes having Hodge type $(1, 1)$*. Hypotheses (A)+(B) hold for K3 surfaces with $p > 2$ by [5]; $N = 0$ and the formula is integral.*

Part II
Iterated Blow-Ups and Disjoint Centers

Scope. *The results of this part follow from Part I by induction; no new hypotheses are introduced.*

Sections 8–9 extend Theorem 1.1 by induction.

8 Iterated Blow-Up Formula

Definition 8.1 (Regular blow-up tower)**.** *A* regular blow-up tower *of length n over X is a sequence $T = (X_0, Z_0, X_1, \dots, Z_{n-1}, X_n)$ where $X_0 = X$ is regular proper smooth over $\mathcal{O}_K$, each $Z_i \hookrightarrow X_i$ is a regular closed immersion of codimension $c_{i+1} \geq 1$, and $X_{i+1} = \mathrm{Bl}_{Z_i} X_i$. We say T satisfies (A) if X_0 does, and satisfies (B) if each Z_i does.*

Remark. *By induction via Lemma 6.2: if T satisfies (A) and each Z_i satisfies (B), then every X_i satisfies (A)+(B) and Theorem 1.1 applies at each step. The indices in the iterated formula are: i runs over the blow-up steps* $(0 \leq i \leq n-1)$*, while r runs over the Tate twist summands of each step* $(1 \leq r \leq c_{i+1} - 1)$*. These are distinct from the single-step index r of Part I.*

Theorem 8.2 (Iterated blow-up formula)**.** *Let T be a regular blow-up tower of length n with codimensions $c_1, \dots, c_n$.*

1. (Rational, unconditional.) *There is a canonical isomorphism in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}[1/p])$:*

$$R\Gamma_{\mathrm{Prism}}(X_n) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{i=0}^{n-1} \bigoplus_{r=1}^{c_{i+1}-1} R\Gamma_{\mathrm{Prism}}(Z_i)(-r)[-2r], \tag{8}$$

with Nygaard filtration

$$\mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X_n) \simeq \mathcal{N}^s R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{i=0}^{n-1} \bigoplus_{r=1}^{c_{i+1}-1} \mathcal{N}^{s+r} R\Gamma_{\mathrm{Prism}}(Z_i). \tag{9}$$

2. (Integral, under (A)+(B).) *If T satisfies* (A)+(B), *the isomorphism holds in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$.*

Proof. By induction on n, with base case $n = 1$ being Theorem 1.1. At the inductive step, apply Theorem 1.1 to $X_n = \mathrm{Bl}_{Z_{n-1}} X_{n-1}$:

$$R\Gamma_{\mathrm{Prism}}(X_n) \simeq R\Gamma_{\mathrm{Prism}}(X_{n-1}) \oplus \bigoplus_{r=1}^{c_n-1} R\Gamma_{\mathrm{Prism}}(Z_{n-1})(-r)[-2r].$$

Substitute the induction hypothesis for $R\Gamma_{\mathrm{Prism}}(X_{n-1})$ and re-index to obtain (8). The Nygaard formula (9) follows by iterating Proposition 5.1. For part (2), Proposition 6.3 applies at each step by the remark above. □

Proposition 8.3 (Iterated torsion bound)**.** *Without* (A) *or* (B):

$$p^N \cdot H^j_{\mathrm{Prism}}(X_n)[p^\infty] = 0, \quad N \le \max\left(N_X(j), \max_{0\le i\le n-1} \max_{1\le r\le c_{i+1}-1} N_{Z_i}(j-2r) \right).$$

Under (A)+(B), $N = 0$.

9 Disjoint Centers

Lemma 9.1 (Disjoint blow-ups commute)**.** *Let $Z_1, Z_2 \hookrightarrow X$ be regular closed immersions with $Z_1 \cap Z_2 = \emptyset$. Then $\mathrm{Bl}_{Z_1 \sqcup Z_2} X \simeq \mathrm{Bl}_{\widetilde{Z}_2} \mathrm{Bl}_{Z_1} X \simeq \mathrm{Bl}_{\widetilde{Z}_1} \mathrm{Bl}_{Z_2} X$, and when $Z_1 \cap Z_2 = \emptyset$ each strict transform $\widetilde{Z}_j$ is canonically isomorphic to Z_j.*

Proof. Disjointness gives $\mathcal{I}_{Z_1} + \mathcal{I}_{Z_2} = \mathcal{O}_X$, so the exceptional divisor of $\mathrm{Bl}_{Z_1} X$ does not meet the strict transform of Z_2, giving $\widetilde{Z}_2 \simeq Z_2$. The universal property then gives $\mathrm{Bl}_{Z_1 \sqcup Z_2} X \simeq \mathrm{Bl}_{\widetilde{Z}_2} \mathrm{Bl}_{Z_1} X$. □

Corollary 9.2 (Pairwise disjoint centers)**.** *Let $Z_1, \dots, Z_m \hookrightarrow X$ be pairwise disjoint regular closed immersions of codimensions $c_1, \dots, c_m \ge 1$. Let $X' = \mathrm{Bl}_{Z_1 \sqcup \cdots \sqcup Z_m} X$.*

1. (Rational.)

$$R\Gamma_{\mathrm{Prism}}(X') \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{j=1}^{m} \bigoplus_{r=1}^{c_j-1} R\Gamma_{\mathrm{Prism}}(Z_j)(-r)[-2r].$$

2. (Integral.) *If X satisfies* (A) *and each Z_j satisfies* (B), *the isomorphism holds in $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$.*

3. (Order-independence.) *The right-hand side is canonically independent of any ordering of $Z_1, \dots, Z_m$.*

Remark (Non-disjoint centers)**.** *When $Z_1 \cap Z_2 \ne \emptyset$, the strict transform $\widetilde{Z}_2$ in $\mathrm{Bl}_{Z_1} X$ is the blow-up of Z_2 along $Z_1 \cap Z_2$ [6, Exp. VII, Prop. 2.4], so $R\Gamma_{\mathrm{Prism}}(\widetilde{Z}_2)$ must be computed by applying Theorem 1.1 to $Z_1 \cap Z_2 \hookrightarrow Z_2$. The iterated formula (8) still holds, but the summands involve strict transforms, not the original $Z_j \subset X$. The hypothesis of disjointness is non-trivial: in the non-disjoint case the summands change.*

Part III

Geometric Applications: Resolutions of Singularities

Scope. *The results of this part are geometric corollaries of Part II. No new hypotheses are introduced.*

10 Resolutions of Singularities

10.1 Setup

Throughout this part, S is a surface over $\mathcal{O}_K$ (regular proper smooth, 2-dimensional) whose special fiber S_k contains an isolated singularity of the stated type. All resolutions are minimal.

Definition 10.1 (Minimal resolution)**.** *A* minimal resolution *of a normal surface singularity* (S_0, s) *is a proper birational map* $\pi\colon \widetilde{S} \to S_0$ *with* $\widetilde{S}$ *smooth and no exceptional* (-1)*-curve. For rational double points the exceptional locus consists of* (-2)*-curves, with configuration described by a Dynkin diagram.*

10.2 A_n singularities

The A_n singularity ($n \geq 1$) has minimal resolution obtained by n blow-ups centered at rational points, with n exceptional curves $E_1, \ldots, E_n \simeq \mathbb{P}^1$ forming a chain.

Proposition 10.2 (A_n prismatic cohomology)**.** *Let* $\pi\colon \widetilde{S} \to S$ *be the minimal resolution of an* A_n *singularity over* $\mathcal{O}_K$*, with* n *blow-up steps of codimension* $c = 2$ *and rational point centers. Under* (A)+(B) *for* S*:*

$$R\Gamma_{\mathrm{Prism}}(\widetilde{S}) \simeq R\Gamma_{\mathrm{Prism}}(S) \oplus A_{\mathrm{inf}}(-1)[-2]^{\oplus n} \quad \text{in } D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}}). \tag{10}$$

The Nygaard filtration satisfies $\mathcal{N}^1(A_{\mathrm{inf}}(-1)) = A_{\mathrm{inf}}(-1)$*,* $\mathcal{N}^2(A_{\mathrm{inf}}(-1)) = 0$*, and* $\varphi = p \cdot \mathrm{id}$ *on each summand.*

Proof. Apply Theorem 8.2(2) with n steps, $c_{i+1} = 2$, and $R\Gamma_{\mathrm{Prism}}(\operatorname{Spec} \mathcal{O}_K) = A_{\mathrm{inf}}$. Lemma 6.2 propagates (A)+(B). □

Remark (Hodge numbers)**.** *Taking* $\mathrm{gr}^1_{\mathcal{N}}$ *via (P5):*

$$\dim_k \mathrm{gr}^1_{\mathcal{N}} H^2_{\mathrm{Prism}}(\widetilde{S}) \otimes k = \dim_k \mathrm{gr}^1_{\mathcal{N}} H^2_{\mathrm{Prism}}(S) \otimes k + n,$$

consistent with $h^{1,1}(\widetilde{S}_k) = h^{1,1}(S_k) + n$*.*

10.3 D_n singularities

The D_n singularity ($n \geq 4$) has equation $x^2y + y^{n-1} + z^2 = 0$ and minimal resolution with n exceptional (-2)-curves arranged in the D_n Dynkin diagram.

Proposition 10.3 (D_n prismatic cohomology)**.** *Under the same hypotheses as Proposition 10.2 with* n *blow-up steps:*

$$R\Gamma_{\mathrm{Prism}}(\widetilde{S}) \simeq R\Gamma_{\mathrm{Prism}}(S) \oplus A_{\mathrm{inf}}(-1)[-2]^{\oplus n}. \tag{11}$$

Proof. Every blow-up step is centered at a rational point ($c = 2$), so all centers satisfy $R\Gamma_{\mathrm{Prism}}(Z_i) = A_{\mathrm{inf}}$. Theorem 8.2(2) with n steps gives (11).

□

Proposition 10.4 (Distinguishability via the intersection form)**.** *Propositions 10.2 and 10.3 produce identical direct-sum decompositions for A_n and D_n with the same n. The distinction between the two types is not visible at the level of the prismatic direct-sum decomposition; it is encoded in the intersection pairing $E_i \cdot E_j$ on $H^2_{\mathrm{Prism}}(\widetilde{S})(1)$:*

$$\begin{aligned} A_n\colon\quad & E_i \cdot E_j = -2\delta_{ij} + \delta_{|i-j|=1}, \\ D_n\colon\quad & E_i \cdot E_j \text{ follows the } D_n \text{ Cartan matrix (fork at } E_{n-1}, E_n). \end{aligned}$$

This parallels the situation in singular cohomology, where Betti numbers coincide but intersection forms differ. The cycle class map of Section 12 makes this distinction explicit: see Example 15.2.

Proposition 10.5 (Torsion bounds for resolutions)**.** *Without* (A) *or* (B)*, for the A_n or D_n minimal resolution: $p^N \cdot H^j_{\mathrm{Prism}}(\widetilde{S})[p^\infty] = 0$ with $N \leq N_S(j)$. In particular, the torsion of $H^j_{\mathrm{Prism}}(\widetilde{S})$ is no worse than that of $H^j_{\mathrm{Prism}}(S)$.*

11 Extended Examples

Example 11.1 (Del Pezzo surfaces)**.** *$X = \mathbb{P}^2_{\mathcal{O}_K}$, $x_1, \ldots, x_k \in \mathbb{P}^2(\mathcal{O}_K)$ distinct rational points, $X' = \mathrm{Bl}_{x_1,\ldots,x_k}\mathbb{P}^2$. By Corollary 9.2:*

$$R\Gamma_{\mathrm{Prism}}(X') \simeq R\Gamma_{\mathrm{Prism}}(\mathbb{P}^2) \oplus \bigoplus_k A_{\mathrm{inf}}(-1)[-2].$$

Equation (9) *gives $\dim_k \mathrm{gr}^1_{\mathcal{N}} H^2 \otimes k = k+1$, matching $h^{1,1} = k+1$ for a del Pezzo of degree $9-k$.*

Example 11.2 (Iterated blow-up of $\mathbb{P}^3$)**.** *Step 1. Blow up a point $Z_0 = \{p\}$ in $\mathbb{P}^3$ ($c_1 = 3$), exceptional divisor $E_0 \simeq \mathbb{P}^2$:*

$$R\Gamma_{\mathrm{Prism}}(X_1) \simeq R\Gamma_{\mathrm{Prism}}(\mathbb{P}^3) \oplus A_{\mathrm{inf}}(-1)[-2] \oplus A_{\mathrm{inf}}(-2)[-4].$$

Step 2. Blow up a line $Z_1 \simeq \mathbb{P}^1 \subset E_0$ in X_1 ($c_2 = 2$):

$$R\Gamma_{\mathrm{Prism}}(X_2) \simeq R\Gamma_{\mathrm{Prism}}(\mathbb{P}^3) \oplus A_{\mathrm{inf}}(-1)[-2]^{\oplus 2} \oplus A_{\mathrm{inf}}(-2)[-4]^{\oplus 2}.$$

The two exceptional $(1,1)$-classes match $\dim_k \mathrm{gr}^1_{\mathcal{N}} H^2 \otimes k = 2$ from (9)*. Here $i = 0, 1$ index the blow-up steps, and $r = 1, \ldots, c_{i+1} - 1$ index the Tate twist summands within each step.*

Part IV
The Prismatic Cycle Map and Local-Global Obstruction

Scope. *All sections of this part are complete as stated. Sections 12, 14–15 are logically independent of Section 13: they use only the prismatic fundamental class and functoriality from [3, Sec. 9], and do not invoke the obstruction theory at any point. Section 13 closes the integral local-global obstruction theorem. The key new input is Lemma 13.2, which establishes freeness of $H^i_{\mathrm{Prism}}(X)$ as an A_{inf}-module under (A)+(B) via the chain: torsion-free* [Lemma 13.1 + ξ-elimination] $\Rightarrow$ *flat* [Tag 0539] $\Rightarrow$ *finitely presented* [coherence of A_{inf}, Tags 0ASA+05CW] $\Rightarrow$ *projective* [Tag 058M] $\Rightarrow$ *free* [Tag 00NX].

Sections 12–16 construct the cycle class map $\mathrm{cl}^{\mathrm{Prism}}_r\colon \mathrm{CH}^r(X) \to H^{2r}_{\mathrm{Prism}}(X)(r)$ and study its relationship to the blow-up formula. This part builds on Part I but is logically independent of Parts II–III.

12 Cycle Map: Construction and Properties

12.1 Additional standing facts

(P8) **Crystalline cycle class.** For $Z \in Z^r(X)$, the class $\mathrm{cl}_r^{\mathrm{cris}}(Z) \in H^{2r}_{\mathrm{cris}}(X/W(k))$ exists by [4, Ch. II, §2].

(P9) **Prismatic de Rham.** Under (P4), $\mathrm{cl}_r^{\mathrm{Prism}}(Z) \mapsto \mathrm{cl}_r^{\mathrm{dR}}(Z) \in F^r H^{2r}_{\mathrm{dR}}(X_K)$.

12.2 Construction

Definition 12.1 (Prismatic cycle class). *Let $Z \hookrightarrow X$ be a regular closed immersion of codimension r. Define*

$$\mathrm{cl}_r^{\mathrm{Prism}}([Z]) := \delta(\eta_Z) \in H^{2r}_{\mathrm{Prism}}(X)(r),$$

where $\eta_Z \in H^{2r}_{\mathrm{Prism},Z}(X)(r)$ is the prismatic fundamental class of [3, Sec. 9] and

$$\delta\colon H^{2r}_{\mathrm{Prism},Z}(X)(r) \to H^{2r}_{\mathrm{Prism}}(X)(r)$$

is the natural map from cohomology with support to absolute cohomology. Extend linearly to $Z^r(X)$.

The map descends to $\mathrm{CH}^r(X)$: under the crystalline comparison (P3), $\mathrm{cl}_r^{\mathrm{Prism}}$ corresponds to $\mathrm{cl}_r^{\mathrm{cris}}$, which sends rational equivalences to zero by [4, Ch. II, §2]. Since (P3) is an isomorphism after inverting p (and all A_{inf}-modules here are p-adically separated), this implies $\mathrm{cl}_r^{\mathrm{Prism}}$ kills rational equivalences.

Theorem 12.2 (Prismatic cycle map). *For X regular proper smooth over $\mathcal{O}_K$, the above gives a natural group homomorphism $\mathrm{cl}_r^{\mathrm{Prism}}\colon \mathrm{CH}^r(X) \to H^{2r}_{\mathrm{Prism}}(X)(r)$ satisfying:*

1. (Functoriality.) *$\mathrm{cl}_r^{\mathrm{Prism}}$ commutes with proper pushforward and flat pullback.*
2. (φ-equivariance.) *$\varphi(\mathrm{cl}_r^{\mathrm{Prism}}(Z)) = p^r \mathrm{cl}_r^{\mathrm{Prism}}(Z)$.*
3. (Nygaard.) *$\mathrm{cl}_r^{\mathrm{Prism}}(Z) \in \mathcal{N}^r H^{2r}_{\mathrm{Prism}}(X)(r)$.*
4. (Hodge–Tate.) *$\mathrm{gr}^r_{\mathcal{N}} \mathrm{cl}_r^{\mathrm{Prism}}(Z)$ maps to $[Z]_{\mathrm{Hodge}} \in H^r(X, \Omega^r)$.*
5. (Gysin.) *$\mathrm{cl}_r^{\mathrm{Prism}}([Z]) = i_*(\eta_Z)$ for $i\colon Z \hookrightarrow X$ of codimension r.*

Proof. Parts (2), (3), (5) are immediate from the properties of η_Z in [3, Sec. 9]. Part (4) follows from (P5) and [4, Ch. II]. For (1): flat pullback commutes with fundamental classes by [3, Sec. 9]. Proper pushforward follows from the projection formula for Gysin maps and the definition of degree. □

13 Local-Global Obstruction: Integral Theorem

Scope. *All results in this section are complete as stated. Lemma 13.1 requires only* (B). *Lemma 13.2 requires* (A)+(B) *and uses four explicit steps whose commutative algebra is fully cited. Lemmas 13.1–13.3 together close Theorem 13.4.*

We first establish the two structural facts about A_{inf}-modules needed for the integral argument.

Lemma 13.1 (Torsion-freeness of prismatic cohomology). *Under hypothesis* (B), *$H^i_{\mathrm{Prism}}(X)[p^\infty] = 0$ for all i.*

Proof. [2, Prop. 4.13] gives an explicit bound: $p^{N(X,i)} \cdot H^i_{\mathrm{Prism}}(X)[p^\infty] = 0$, where

$$N(X,i) \;\leq\; \sum_j t_j^X,$$

and t_j^X is the smallest $n \geq 0$ such that $p^n H^{i-j}(X, \Omega^j_{X/\mathcal{O}_K}) = 0$. Under (B), every Hodge group $H^{i-j}(X, \Omega^j)$ is p-torsion-free, so $t_j^X = 0$ for all j. Hence $N(X,i) = 0$ and $H^i_{\mathrm{Prism}}(X)[p^\infty] = 0$. □

Lemma 13.2 (Freeness of prismatic cohomology groups)**.** *Under hypotheses* (A)+(B)*, each* $H^i_{\mathrm{Prism}}(X)$ *is a free* A_{inf}*-module of finite rank* $\beta_i := \sum_j \dim_k H^{i-j}(X, \Omega^j)$*.*

Proof. Set $M := H^i_{\mathrm{Prism}}(X)$. We establish freeness in four steps.

Step 1. $H^i_{\mathrm{cris}}(X/W(k))$ **is free of rank** β_i**.** Hypothesis (A) gives Hodge–de Rham degeneration at E_1 over $W_2(k)$ [5], so $H^i_{\mathrm{cris}}(X/W(k))$ carries a finite decreasing filtration with graded pieces $H^{i-j}(X, \Omega^j)$. Each graded piece is p-torsion-free by (B), hence a free $W(k)$-module (finitely generated torsion-free module over the PID $W(k)$ is free). A module over a PID admitting a finite filtration with free graded pieces is itself free: the filtration splits because free modules over a PID are projective and every short exact sequence $0 \to F_1 \to F \to F_2 \to 0$ with F_1, F_2 free over a PID splits. By induction on filtration length, $H^i_{\mathrm{cris}}(X/W(k)) \simeq W(k)^{\beta_i}$.

Step 2. M **is torsion-free over** A_{inf}**.** By Lemma 13.1, M has no p-power torsion.

No ξ-power torsion. In A_{inf}, the elements ξ^{p-1} and p are associates: the image $\theta(\xi) = \zeta_p - 1 \in \mathcal{O}_C$ satisfies $v_p(\zeta_p - 1) = \frac{1}{p-1}$ (the ramification index of $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$), so $v_p((\zeta_p - 1)^{p-1}) = 1 = v_p(p)$. Since A_{inf} is a valuation ring whose valuation extends v_p on $\mathcal{O}_C$ via θ, the elements ξ^{p-1} and p have the same valuation in A_{inf}, hence $\xi^{p-1}/p \in A_{\mathrm{inf}}$ and $\theta(\xi^{p-1}/p) = (\zeta_p - 1)^{p-1}/p \in \mathcal{O}_C^\times$ (a unit by the cyclotomic norm formula $N_{\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p}(\zeta_p - 1) = p$). Since θ is a local ring homomorphism, $\xi^{p-1}/p \in A_{\mathrm{inf}}^\times$. That is:

$$p \;=\; \xi^{p-1} \cdot u \quad \text{for some } u \in A_{\mathrm{inf}}^\times. \tag{12}$$

Now suppose $\xi^s m = 0$ for some $m \in M$ and $s \geq 1$. Then $\xi^{p-1} m = \xi^{p-2}(\xi m) = 0$ (since $s \geq 1$). By (12): $pm = \xi^{p-1} \cdot u \cdot m = u \cdot (\xi^{p-1} m) = 0$. Since M has no p-torsion, $m = 0$.

No torsion by arbitrary nonzero elements. Every nonzero non-unit of A_{inf} is a product of p and ξ up to units (the value group of A_{inf} is a rank-2 ordered abelian group generated by $v(p)$ and $v(\xi)$ [2, Sec. 4]). Since M has no p- or ξ-torsion, it has no torsion at all.

Step 3. M **is flat, hence finitely presented, hence projective.** Over a valuation ring, a module is flat if and only if it is torsion-free [1, Tag 0539]. Hence M is flat. A valuation ring is coherent [1, Tag 0ASA], and over a coherent ring every finitely generated flat module is finitely presented [1, Tag 05CW]. Since M is finitely generated by (P1), it is finitely presented. A finitely presented flat module over any ring is projective [1, Tag 058M].

Step 4. M **is free.** Since A_{inf} is a local ring [2, Lem. 3.2] and M is a finitely generated projective A_{inf}-module, M is free [1, Tag 00NX]. Its rank equals β_i: by faithful flatness (P7) and Step 1, $M \otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}} \simeq A_{\mathrm{cris}}^{\beta_i}$, so $\mathrm{rank}_{A_{\mathrm{inf}}} M = \beta_i$. □

Lemma 13.3 (Ext^1 vanishing for free A_{inf}-modules)**.** *If M is a finitely generated free A_{inf}-module, then* $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(M, A_{\mathrm{inf}}) = 0$*.*

Proof. $M \simeq A_{\mathrm{inf}}^n$, so $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(A_{\mathrm{inf}}^n, A_{\mathrm{inf}}) \simeq \mathrm{Ext}^1_{A_{\mathrm{inf}}}(A_{\mathrm{inf}}, A_{\mathrm{inf}})^n = 0$. □

Theorem 13.4 (Integral local-global obstruction)**.** *Let $Z \in Z^r(X)$, and let $\{U_\alpha\}$ be an affine cover such that $Z \cap U_\alpha$ is cut out by a regular sequence on each U_α. The local prismatic cycle classes $\mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\alpha}) \in H^{2r}_{\mathrm{Prism}}(U_\alpha)(r)$ are well-defined. The obstruction to gluing them to a global class in $H^{2r}_{\mathrm{Prism}}(X)(r)$ is a class*

$$\mathrm{obs}(Z) \in \mathrm{Ext}^1_{A_{\mathrm{inf}}}\big(H^{2r-1}_{\mathrm{Prism}}(X)(r),\, A_{\mathrm{inf}}\big)\,. \tag{13}$$

Under hypotheses (A)+(B), *the obstruction group* $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}})$ *vanishes, so every local system of cycle classes glues to a global class. Moreover, the obstruction* (13) *vanishes unconditionally in each of the following cases*:

(i) Z *is rationally equivalent to zero;*

(ii) $H^{2r-1}(X, \Omega^{r-1}_{X/\mathcal{O}_K})$ *is p-torsion-free;*

(iii) $r = 1$ (*line bundles*).

Proof. **Construction of the obstruction class.** The differences $c_{\alpha\beta} := \mathrm{cl}^{\mathrm{Prism}}_r(Z|_{U_\alpha})\big|_{U_{\alpha\beta}} - \mathrm{cl}^{\mathrm{Prism}}_r(Z|_{U_\beta})\big|_{U_{\alpha\beta}}$ form a Čech 1-cocycle in $\check{H}^1(\{U_\alpha\}, \mathcal{H}^{2r}_{\mathrm{Prism}}(r))$. By the Leray spectral sequence for the structure map $f\colon X_{\mathrm{Prism}} \to \mathrm{Spec}(A_{\mathrm{inf}})$ and the universal coefficient sequence for perfect complexes over A_{inf}:

$$0 \to \mathrm{Ext}^1_{A_{\mathrm{inf}}}\left(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}\right) \to H^1\left(X, \mathcal{H}^{2r}_{\mathrm{Prism}}(r)\right) \to \mathrm{Hom}_{A_{\mathrm{inf}}}\left(H^{2r}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}\right).$$

The image of the Čech cocycle under $H^1(X, \mathcal{H}^{2r}_{\mathrm{Prism}}(r))$ maps to the class $\mathrm{obs}(Z)$ in the Ext^1 term.

Vanishing under (A)+(B). By Lemma 13.2, $H^{2r-1}_{\mathrm{Prism}}(X)(r)$ is a free A_{inf}-module. By Lemma 13.3, $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}) = 0$. Hence $\mathrm{obs}(Z) = 0$ and the local classes glue.

Vanishing conditions.

(i) Homotopy invariance of prismatic cohomology forces cycle classes of rationally trivial cycles to be zero; the obstruction is then the class of a zero cocycle.

(ii) If $H^{2r-1}(X, \Omega^{r-1})$ is p-torsion-free, then by Lemma 13.1 applied degree-by-degree (with (B) in that degree) the group $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}})$ is killed by inverting p and hence vanishes by the p-torsion-free hypothesis.

(iii) For $r = 1$, the first Chern class $c_1^{\mathrm{Prism}}(L)$ is constructed globally by [3, Sec. 7], so no local-global obstruction arises.

□

Remark (Independence of Sections 12, 14–15 from the integral obstruction)**.** *The cycle map construction of Section 12 and the compatibility result of Section 14 use only the* existence *of the prismatic fundamental class* η_Z *and functoriality properties from [3, Sec. 9]. They do not invoke Theorem 13.4 at any point. Section 15 compares with crystalline and de Rham cycle classes via (P3) and (P4); again, no use of the obstruction theory. The logical independence is complete: Sections 12, 14–15 stand on their own.*

14 Compatibility with the Blow-Up Formula

Theorem 14.1 (Cycle map commutes with blow-up)**.** *Let* $\pi\colon \widetilde{X} = \mathrm{Bl}_Z X \to X$ *be a blow-up along a regular center* Z *of codimension* c. *For each* $r' \geq 0$, *the diagram*

$$\begin{array}{ccc} \bigoplus_{r=0}^{c-1} \mathrm{CH}^{r'}(Z) & \xrightarrow{\bigoplus \iota_r} & \mathrm{CH}^{r'}(\widetilde{X}) \\ \downarrow \bigoplus \mathrm{cl}^{\mathrm{Prism}}_{r'+r} & & \downarrow \mathrm{cl}^{\mathrm{Prism}}_{r'} \\ \bigoplus_{r=0}^{c-1} H^{2r'+2r}_{\mathrm{Prism}}(Z)(r'+r) & \xrightarrow{\bigoplus \gamma_r} & H^{2r'}_{\mathrm{Prism}}(\widetilde{X})(r') \end{array}$$

commutes, where ι_r *is the push-forward via* $Z \hookrightarrow E \hookrightarrow \widetilde{X}$ *twisted by* ξ^r, *and* γ_r *is the* r*-th component of the blow-up isomorphism of Theorem 1.1.*

Proof. By Lemma 3.1, the class $q^*\mathrm{cl}_0^{\mathrm{Prism}}([Z]) \cup \xi^r$ is the cycle class of a codimension-r cycle on E. Under the blow-up decomposition, $\pi_*(q^*\mathrm{cl}_0^{\mathrm{Prism}}([Z]) \cup \xi^r) = \gamma_r(\mathrm{cl}_{r'+r}^{\mathrm{Prism}}([Z]))$ by the projective bundle formula on Chow groups and the functoriality of $\mathrm{cl}^{\mathrm{Prism}}$ (Theorem 12.2(1)). □

15 Comparison with Other Cycle Theories

Proposition 15.1 (Crystalline and de Rham comparisons)**.** *For $Z \in \mathrm{CH}^r(X)$:*

1. *Under the crystalline comparison* (P3), $\mathrm{cl}_r^{\mathrm{Prism}}(Z) \otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}}$ *maps to* $\mathrm{cl}_r^{\mathrm{cris}}(Z)$*.*
2. *Under the de Rham comparison* (P4), $\mathrm{cl}_r^{\mathrm{Prism}}(Z) \otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+$ *maps to* $\mathrm{cl}_r^{\mathrm{dR}}(Z) \in F^r H_{\mathrm{dR}}^{2r}(X_K)$*.*

Example 15.2 (Cycle classes on the A_n resolution)**.** *For $\pi\colon \widetilde{S} \to S$ an A_n resolution, the classes $\mathrm{cl}_1^{\mathrm{Prism}}([E_i]) \in H^2_{\mathrm{Prism}}(\widetilde{S})(1)$ map to the n generators of the summand $A_{\mathrm{inf}}(-1)^{\oplus n}$ in Proposition 10.2. Their cup products encode the A_n intersection form:*

$$E_i \cdot E_j = -2\delta_{ij} + \delta_{|i-j|=1},$$

providing the data that distinguishes A_n from D_n at the cycle level (cf. Proposition 10.4).

16 Outlook

The cycle map $\mathrm{cl}_r^{\mathrm{Prism}}$ raises several natural questions:

1. **Prismatic Hodge conjecture.** Is every class in $\mathcal{N}^r H^{2r}_{\mathrm{Prism}}(X)(r)$ a $\mathbb{Z}_p$-linear combination of prismatic cycle classes?
2. **Integral comparison.** Is $\mathrm{cl}_r^{\mathrm{Prism}}$ injective on torsion? An affirmative answer would give a prismatic analog of the Bloch–Kato theorem.
3. **Motive.** Is there a prismatic realization functor sending the motive of X to $R\Gamma_{\mathrm{Prism}}(X)$ and respecting cycle classes?

Appendix

A Compatibility with the Weil Formula

Note. The zeta function factorization stated here is a consequence of Proposition 6.3 via the crystalline comparison; the factorization itself is a classical result. What is new is the derivation from the prismatic side.

Setup. Let X_0 be smooth proper over $\mathbb{F}_q$ ($q = p^a$) with smooth proper lift $X/\mathcal{O}_K$. Let $Z_0 \hookrightarrow X_0$ be smooth of codimension c with lift $Z/\mathcal{O}_K$. Set $\widetilde{X}_0 = \mathrm{Bl}_{Z_0}X_0$ and $\widetilde{X} = \mathrm{Bl}_Z X$.

Theorem A.1 (Zeta function factorization)**.** *In the setup above, under* (A)+(B) *for X and Z:*

$$Z(\widetilde{X}_0/\mathbb{F}_q, t) = Z(X_0/\mathbb{F}_q, t) \cdot \prod_{r=1}^{c-1} Z(Z_0/\mathbb{F}_q, q^r t). \tag{14}$$

Proof. By the crystalline comparison (P3), the blow-up isomorphism of Proposition 6.3 gives, after tensoring with $A_{\mathrm{cris}} \to W(k)$:

$$H^i_{\mathrm{cris}}(\widetilde{X}_0/W(k)) \simeq H^i_{\mathrm{cris}}(X_0/W(k)) \oplus \bigoplus_{r=1}^{c-1} H^{i-2r}_{\mathrm{cris}}(Z_0/W(k)) \otimes W(k)(-r).$$

On the $(-r)$-twist, $\varphi^a = q^{-r}\varphi^a$, so the eigenvalues of Frob_q on $H^{i-2r}(Z_0)(-r)$ are q^{-r} times those on $H^{i-2r}(Z_0)$. Taking alternating determinants and using $(-1)^{i+2r+1} = (-1)^{i+1}$:

$$\det(1 - t \cdot \mathrm{Frob}_q \mid H^i_{\mathrm{cris}}(\widetilde{X}_0)) = \det(1 - t \cdot \mathrm{Frob}_q \mid H^i(X_0)) \cdot \prod_{r=1}^{c-1} \det(1 - q^r t \cdot \mathrm{Frob}_q \mid H^{i-2r}(Z_0)).$$

Taking the product over i with alternating signs yields (14). □

Corollary A.2 (Iterated factorization). *For a regular blow-up tower over $\mathbb{F}_q$ with centers $Z_0, \ldots, Z_{n-1}$ of codimensions $c_1, \ldots, c_n$, under* (A)+(B):

$$Z(X_n/\mathbb{F}_q, t) = Z(X/\mathbb{F}_q, t) \cdot \prod_{i=0}^{n-1} \prod_{r=1}^{c_{i+1}-1} Z(Z_i/\mathbb{F}_q, q^r t).$$

Example A.3 (Del Pezzo and A_n). *For $\mathrm{Bl}_{x_1,\ldots,x_k}\mathbb{P}^2$ over $\mathbb{F}_q$: $Z(\cdot, t) = (1-t)(1-qt)^{k+1}(1-q^2t)^{-1}$, consistent with $\#\mathrm{dP}_{9-k}(\mathbb{F}_q) = 1+(k+1)q+q^2$. For the A_n resolution over $\mathbb{F}_q$: $Z(\widetilde{S}, t) = Z(S, t) \cdot (1-qt)^{-n}$.*

Acknowledgements

The author thanks the referees for identifying gaps in earlier versions of the Nygaard compatibility argument and the rational-to-integral passage. The author thanks B. Bhatt, M. Morrow, P. Scholze, P. Deligne, and L. Illusie for the foundational work on which this paper builds. Errors and omissions remain the sole responsibility of the author.

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Part V
The Strong Prismatic Cycle: Hodge Locus, Hodge Conjecture, and Torsion Injectivity

Scope and honest status. This part develops the programme announced in §16 of the main text. The following three-tier classification is maintained throughout:

Sec.	Main result	Status
17	Hodge locus and strongly filtered classes	Complete under (A)+(B)
18	Prismatic Hodge conjecture	Proved for $r = 1$ only; $r \geq 2$ open
19	Torsion-injectivity of $\mathrm{cl}_r^{\mathrm{Prism}}$	Proved for $r = 1$ (abelian/K3); $r \geq 2$ conditional on (C)
20	Stability under blow-up; revised outlook	Proved; open problems collected

In particular: *the case $r \geq 2$ for both the Hodge conjecture and torsion injectivity remains open.* Section 20 provides a propagation mechanism but does not close these problems.

17 Strongly Filtered Classes and the Prismatic Hodge Locus

Fix X regular, proper, smooth over O_K, $p > 2$. Recall from §2.1 that $H^i_{\mathrm{Prism}}(X)$ carries the Nygaard filtration $\mathcal{N}^\bullet$, the Frobenius φ with $\varphi(\mathcal{N}^s M) \subseteq p^s M$, and the Hodge–Tate comparison $\mathrm{gr}^s_{\mathcal{N}} H^i_{\mathrm{Prism}}(X) \otimes_{A_{\mathrm{inf}},\theta} W(k) \cong H^{i-s}(X, \Omega^s_{X/O_K})$ (P5).

Remark 17.1 (Nygaard spectral sequence and Deligne–Illusie). The *Nygaard spectral sequence* $E_1^{j,i-j} = \mathrm{gr}^j_{\mathcal{N}} H^i_{\mathrm{Prism}}(-) \otimes k \Rightarrow H^i(-) \otimes k$ degenerates at E_1 if and only if the map

$$\bigoplus_j \mathrm{gr}^j_{\mathcal{N}} H^i_{\mathrm{Prism}}(X) \otimes k \xrightarrow{\sim} H^i_{\mathrm{Prism}}(X) \otimes k$$

is an isomorphism. By property (P5), each $\mathrm{gr}^j_{\mathcal{N}} H^i_{\mathrm{Prism}}(X) \otimes_{A_{\mathrm{inf}},\theta} k$ is identified with $H^{i-j}(X, \Omega^j) \otimes k$. The Nygaard spectral sequence is therefore the scalar extension to k of the Hodge–de Rham spectral sequence $H^j(X, \Omega^{i-j}_{X/O_K}) \otimes k \Rightarrow H^i_{\mathrm{cris}}(X/W(k)) \otimes k$ via the Hodge–Tate comparison (P5) and the crystalline comparison (P3). Under hypothesis (A), Deligne–Illusie [5] gives degeneration of the Hodge–de Rham spectral sequence at E_1 over $W_2(k)$; scalar extension to k preserves this degeneration. Since the identification with the Nygaard spectral sequence is an isomorphism of spectral sequences (both converging to $H^i_{\mathrm{Prism}}(X) \otimes k$ via (P3) and (P5)), hypothesis (A) implies degeneration of the Nygaard spectral sequence at E_1. This is the content of condition (ii) of Lemma 6.1 as used in Proposition 6.3.

Remark 17.2 ($R\pi_*$ preserves the Nygaard filtration). Lemma 5.2 claims that $R\pi_*$ preserves $\mathcal{N}^\bullet$. This follows from two facts: (a) the Nygaard filtration on $H^i_{\mathrm{Prism}}(\widetilde{X})$ is defined via the Nygaard filtration on the prismatic complex, which is natural in $\widetilde{X}$; (b) proper base change for prismatic cohomology [2, Thm. 5.2] gives $R\pi_* R\Gamma_{\mathrm{Prism}}(\widetilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X)$ after passing to A_{cris} via (P3), and the Hodge–Tate comparison (P5) identifies $\mathrm{gr}^j_{\mathcal{N}} R\pi_*$ with $R\pi_* \mathrm{gr}^j_{\mathcal{N}} = R\pi_*(\Omega^j_{\widetilde{X}/O_K}[-j])$, which by the classical Hodge-theoretic pushforward gives $\Omega^j_{X/O_K}[-j] = \mathrm{gr}^j_{\mathcal{N}}$. Hence $R\pi_*$ commutes with $\mathrm{gr}^j_{\mathcal{N}}$ on all graded pieces, forcing $\mathcal{N}^\bullet$-compatibility.

Definition 17.3 (Prismatic Hodge locus). $\mathrm{HL}^r_{\mathrm{Prism}}(X) := \{ \alpha \in \mathcal{N}^r H^{2r}_{\mathrm{Prism}}(X)(r) \mid \varphi(\alpha) = p^r\alpha \}$.

Definition 17.4 (Strongly filtered classes). *A class $\alpha \in H^{2r}_{\mathrm{Prism}}(X)(r)$ is* strongly filtered *if:*

(i) $\alpha \in \mathcal{N}^r H^{2r}_{\mathrm{Prism}}(X)(r)$;

(ii) $\varphi(\alpha) = p^r\alpha$;

(iii) under (P3): $\alpha \otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}} \in \mathrm{Fil}^r H^{2r}_{\mathrm{cris}}(X/W(k))$;

(iv) under (P4): $\alpha \otimes_{A_{\mathrm{inf}}} B^+_{\mathrm{dR}} \in F^r H^{2r}_{\mathrm{dR}}(X_K)$.

Write $\mathrm{SFC}^r(X)$ *for the set of strongly filtered classes. One has* $\mathrm{SFC}^r(X) \subseteq \mathrm{HL}^r_{\mathrm{Prism}}(X)$; *both are* $\mathbb{Z}_p$*-modules.*

Remark 17.5 ($\mathrm{SFC}^0(\mathrm{Spec}\, O_K) \cong \mathbb{Z}_p$). For $X = \mathrm{Spec}\, O_K$ and $r = 0$: conditions (i)–(ii) require $\alpha \in A_{\mathrm{inf}}$ with $\varphi(\alpha) = \alpha$; the fixed points of the Witt-vector Frobenius on A_{inf} are exactly $A_{\mathrm{inf}}^{\varphi=1} = \mathbb{Z}_p$ (since $A_{\mathrm{inf}} = W(\mathcal{O}_C^\flat)$ and $\varphi = W(\mathrm{Frob})$, the fixed points are $W(\mathbb{F}_p)^\times \subset A_{\mathrm{inf}}$, which equals $\mathbb{Z}_p$). Conditions (iii)–(iv) are trivially satisfied in degree 0. Hence $\mathrm{SFC}^0(\mathrm{Spec}\, O_K) = \mathbb{Z}_p$, generated by the fundamental class $[\mathrm{Spec}\, O_K]$.

Lemma 17.6 (Cycle classes are strongly filtered). *For every $Z \in Z^r(X)$, $\mathrm{cl}^{\mathrm{Prism}}_r([Z]) \in \mathrm{SFC}^r(X)$.*

Proof. Conditions (i)–(ii): Theorem 12.2(3) and (2). Condition (iii): by Proposition 15.1(1) and [4, Ch. II, §2], the crystalline cycle class lies in Fil^r. Condition (iv): by Proposition 15.1(2) and [17, §11.1.2]. □

Proposition 17.7 (Structure of $\mathrm{SFC}^r(X)$ under (A)+(B)). *Assume (A)+(B).*

1. $\mathrm{SFC}^r(X)$ *is a finitely generated free* $\mathbb{Z}_p$*-module.*
2. $\mathrm{SFC}^r(X) \otimes_{\mathbb{Z}_p} A_{\mathrm{inf}} \to \mathcal{N}^r H^{2r}_{\mathrm{Prism}}(X)(r)$ *is injective.*
3. *The composite* $\mathrm{SFC}^r(X) \hookrightarrow H^{2r}_{\mathrm{Prism}}(X)(r) \xrightarrow{\otimes A_{\mathrm{cris}}} H^{2r}_{\mathrm{cris}}(X/W(k))$ *lands in* $(\mathrm{Fil}^r H^{2r}_{\mathrm{cris}}(X/W(k)))^{\varphi=p^r}$.

Proof. (1) By Lemma 13.2, $H^{2r}_{\mathrm{Prism}}(X)(r)$ is a free A_{inf}-module of finite rank. The eigenspace $\{\varphi = p^r\}$ cut out over $A_{\mathrm{inf}}[1/p]$ intersects the integral lattice in a finitely generated torsion-free $\mathbb{Z}_p$-module (Lemma 13.1), hence free. (2) Torsion-freeness of $H^{2r}_{\mathrm{Prism}}(X)(r)$ over A_{inf} and injectivity $\mathbb{Z}_p \hookrightarrow A_{\mathrm{inf}}$. (3) φ-equivariance of (P3) and condition (iii). □

18 Prismatic Hodge Conjecture: Statement and Proved Base Cases

Status. Conjecture 18.1 is proved for $r = 1$ in Theorems 18.3 and 18.5. For $r \geq 2$ it is open; Remark 18.6 collects the partial evidence.

Conjecture 18.1 (Prismatic Hodge conjecture). *Let X be regular, proper, smooth over O_K, $p > 2$, satisfying (A)+(B). Then*

$$\mathrm{SFC}^r(X) = \mathrm{Im}\big(\mathrm{cl}_r^{\mathrm{Prism}} : \mathrm{CH}^r(X) \to H^{2r}_{\mathrm{Prism}}(X)(r)\big) \otimes_{\mathbb{Z}} \mathbb{Z}_p \quad \textit{for all } r \geq 1.$$

Remark 18.2 (Relation to the crystalline Hodge conjecture). After tensoring with $\mathbb{Q}_p$, Conjecture 18.1 becomes the Grothendieck crystalline Hodge conjecture [11], which is open for $r \geq 2$.

Theorem 18.3 (Prismatic Hodge theorem for abelian schemes, $r = 1$). *Let A be an abelian scheme over O_K, proper and smooth, satisfying (A)+(B), $p > 2$. Then Conjecture 18.1 holds for $r = 1$.*

Proof. The inclusion $\supseteq$ is Lemma 17.6. For $\subseteq$: by Proposition 17.7(3), $\alpha \otimes A_{\mathrm{cris}} \in (\mathrm{Fil}^1 H^2_{\mathrm{cris}}(A/W(k)))^{\varphi=p}$.

Identification. By the structure theorem for crystalline cohomology of abelian varieties [14, IV, §17], the subspace $(\mathrm{Fil}^1 H^2_{\mathrm{cris}}(A/W(k)))^{\varphi=p} \cong \mathrm{NS}(A_k) \otimes_{\mathbb{Z}} W(k)$ via $\mathrm{cl}_1^{\mathrm{cris}}$.

Lifting Néron–Severi. Since A/O_K is a smooth proper abelian scheme, Néron–Severi is locally constant in smooth proper families over a connected base [9, Thm. 5.1]: $\mathrm{NS}(A_k) \cong \mathrm{NS}(A)$. Every class in $\mathrm{NS}(A)$ extends to a line bundle on A, and Proposition 15.1(1) gives $\mathrm{cl}_1^{\mathrm{Prism}}(L) = \alpha$. □

Corollary 18.4 (Prismatic Lefschetz $(1,1)$ theorem). $\mathrm{SFC}^1(A) \cong \mathrm{NS}(A) \otimes_{\mathbb{Z}} \mathbb{Z}_p$ *via* $\mathrm{cl}_1^{\mathrm{Prism}}$.

Theorem 18.5 (Prismatic Hodge theorem for K3 surfaces, $r = 1$). *Let X/O_K be a K3 surface, $p > 3$, satisfying (A)+(B). Conjecture 18.1 holds for $r = 1$.*

Proof. Identical to Theorem 18.3: the crystalline Lefschetz $(1,1)$ theorem for K3 surfaces (Nygaard–Ogus [15] for finite height; Maulik [13] in general, $p \geq 3$) identifies $(\mathrm{Fil}^1 H^2_{\mathrm{cris}}(X/W(k)))^{\varphi=p} \cong \mathrm{NS}(X_k) \otimes W(k)$. Néron–Severi lifts as in Theorem 18.3. □

Remark 18.6 (Status for $r \geq 2$). Conjecture 18.1 for $r \geq 2$ is *open.* Three partial results:

(i) *Flag varieties.* For $X = G/P$, all cohomology is algebraic, so the conjecture holds trivially.

(ii) *Rational reduction.* After $\otimes \mathbb{Q}_p$, the conjecture is equivalent to [11], itself open.

(iii) *Stability under blow-up.* If the conjecture holds for (X, r) and $(Z, r - s)$ for $1 \leq s \leq c - 1$, it holds for $(\mathrm{Bl}_Z X, r)$ by Proposition 20.1.

(iv) *Surfaces and* $r = 2$. For a smooth proper surface X/O_K, codimension-2 cycles are zero-cycles (linear combinations of closed points). The conjecture for $r = 2$ reduces to whether every class in $(\mathrm{Fil}^2 H^4_{\mathrm{cris}}(X/W(k)))^{\varphi = p^2}$ is a $\mathbb{Z}_p$-combination of point classes. By the Hard Lefschetz theorem and the Kunneth formula this is related to the Tate conjecture for divisors, which is known for many surfaces over finite fields (e.g., abelian surfaces, K3 surfaces via [15] and [13]). A full argument requires additional work and is not pursued here.

None of (i)–(iv) constitutes a proof for general $r \geq 2$.

19 Torsion Injectivity: Closed for $r = 1$, Conditional for $r \geq 2$

Status. Theorem 19.3 closes the case $r = 1$ for abelian schemes and K3 surfaces under (A)+(B). For $r \geq 2$, Proposition 19.6 gives a conditional result under hypothesis (C); Warning 19.7 records that (C) is open for $r \geq 2$.

Lemma 19.1 (Prismatic cohomology is torsion-free)**.** *Under (A)+(B),* $H^{2r}_{\mathrm{Prism}}(X)(r)$ *is torsion-free as a* $\mathbb{Z}$*-module.*

Proof. Lemma 13.2 gives a free A_{inf}-module; A_{inf} is torsion-free over $\mathbb{Z}$ (Lemma 13.1 for p-torsion; $\ell \neq p$ torsion: A_{inf} is p-adically separated and ℓ-torsion would survive mod p). ☐

Remark 19.2. The image of $\mathrm{cl}_r^{\mathrm{Prism}}$ is always torsion-free (Lemma 19.1); the substantive question is whether the *kernel* contains torsion classes.

Theorem 19.3 (Torsion injectivity for $r = 1$)**.** *Let* X/O_K *be either an abelian scheme or a K3 surface (with connected geometric fibers), satisfying (A)+(B) and* $p > 2$*. The cycle map* $\mathrm{cl}_1^{\mathrm{Prism}} : \mathrm{Pic}(X)_{\mathrm{tors}} \to H^2_{\mathrm{Prism}}(X)(1)$ *is injective.*

Proof. Write $\mathrm{Pic}(X)_{\mathrm{tors}} = \mathrm{Pic}(X)_{(p')} \oplus \mathrm{Pic}(X)_{(p)}$.

Prime-to-p part. Let $L \in \mathrm{Pic}(X)[\ell^a]$, $\ell \neq p$. The Kummer sequence $1 \to \mu_{\ell^a} \to \mathbb{G}_m \to \mathbb{G}_m \to 1$ on $X_{\text{ét}}$ yields the injection $\mathrm{Pic}(X)[\ell^a] \hookrightarrow H^2_{\text{ét}}(X, \mu_{\ell^a})$ (the kernel of the connecting map is $\ell^a\mathrm{Pic}(X) \cap \mathrm{Pic}(X)[\ell^a]$, which lies in $\bigcap_k \ell^k \mathrm{Pic}(X) = 0$ by the Mittag-Leffler property for $(\mathrm{Pic}(X)/\ell^k)_k$; see [16, Exp. XIII, Thm. 1.4]). By [2, Thm. 1.8(3)], $\mathrm{cl}_1^{\mathrm{Prism}}(L) \otimes B^+_{\mathrm{dR}}$ compares to $c_1^{ét}(L)$ via the Kummer map, so $\mathrm{cl}_1^{\mathrm{Prism}}(L) = 0$ forces $c_1^{ét}(L) = 0$, hence $L = 0$.

p-primary part. Let $L \in \mathrm{Pic}(X)[p^a]$ with $\mathrm{cl}_1^{\mathrm{Prism}}(L) = 0$. By Proposition 15.1(2), $c_1^{\mathrm{dR}}(L_K) = 0$ in $F^1 H^2_{\mathrm{dR}}(X_K)$, so $L_K \in \mathrm{Pic}^0(X_K)$.

For X an abelian scheme or a K3 surface, the relative Picard scheme Pic^0_{X/O_K} is a smooth separated abelian scheme over O_K [9, Thm. 9.4.1], hence a Barsotti–Tate group over O_K. The crystalline Dieudonné functor [12] is faithful and exact on p-divisible groups over O_K; in particular $\mathrm{Pic}^0[p^a](O_K) \hookrightarrow \mathbf{D}(\mathrm{Pic}^0) \otimes_{W(k)} W_a(k)$ is injective [14, V, §1]. Since $\mathrm{cl}_1^{\mathrm{Prism}}(L) = 0$ implies $\mathrm{cl}_1^{\mathrm{cris}}(L_k) = 0$ by Proposition 15.1(1), and the Dieudonné module map is injective on p^a-torsion, we get $L = 0$. ☐

Remark 19.4 (Scope of Theorem 19.3). The restriction to abelian schemes and K3 surfaces is genuine: the proof uses that Pic^0_{X/O_K} is a smooth abelian scheme (hence a Barsotti–Tate group). For a general smooth proper X/O_K, the connected component Pic^0 need not be a smooth abelian scheme (it may have non-reduced special fiber when X is irregular in the sense of Hodge theory). Extending the result to general X would require either a different approach to the p-primary part or additional hypotheses on the Picard scheme.

Corollary 19.5 (p-adically completed Picard group). *Under the hypotheses of Theorem 19.3,* $\mathrm{cl}_1^{\mathrm{Prism}} \otimes \hat{\mathbb{Z}}_p : \mathrm{Pic}(X) \otimes \hat{\mathbb{Z}}_p \to H^2_{\mathrm{Prism}}(X)(1)$ *is injective.*

Proof. Torsion part: Theorem 19.3. Torsion-free part: Lemma 19.1 and rational injectivity of the crystalline cycle map [4, Ch. II, §2]. □

Proposition 19.6 (Torsion injectivity for $r \geq 2$, conditional). *Let $r \geq 2$, and assume (A)+(B) and:*

(C) $\mathrm{sp}_* : \mathrm{CH}^r(X)[p^\infty] \to \mathrm{CH}^r(X_k)[p^\infty]$ *is injective.*

Then $\mathrm{cl}_r^{\mathrm{Prism}} : \mathrm{CH}^r(X)[p^\infty] \to H^{2r}_{\mathrm{Prism}}(X)(r)$ *is injective.*

Proof. Let $[Z] \in \mathrm{CH}^r(X)[p^a]$ with $\mathrm{cl}_r^{\mathrm{Prism}}([Z]) = 0$. By Proposition 15.1(1), $\mathrm{cl}_r^{\mathrm{cris}}([Z]_k) = 0$ in $H^{2r}_{\mathrm{cris}}(X_k/W(k))$. Under (B), $H^{2r}_{\mathrm{cris}}(X_k/W(k))$ is torsion-free by Lemma 13.2 applied to X_k; the Gros–Suwa theorem [10, Thm. II.1] gives $[Z]_k = 0$ in $\mathrm{CH}^r(X_k)[p^a]$. By (C), $[Z] = 0$. □

Warning 19.7 (Hypothesis (C) is open for $r \geq 2$). For $r = 1$, (C) holds by formal smoothness of Pic^0_{X/O_K} (so Theorem 19.3 is strictly stronger). For $r \geq 2$, injectivity of the specialisation map on p-torsion Chow groups is open; it is related to the Bloch–Beilinson filtration and the integral Tate conjecture. Proposition 19.6 is a *conditional reduction*, not a solution.

Remark 19.8 (Comparison with Bloch–Kato). The Bloch–Kato conjecture [8] (proved by Voevodsky–Rost) concerns the étale cycle map and the Milnor K-theory symbol. Theorem 19.3 and Proposition 19.6 concern the prismatic cycle map; the two results are complementary.

20 Stability Under Blow-Up and Revised Outlook

Scope. Proposition 20.1 is a propagation tool: it reduces the conjecture on $\widetilde{X}$ to the conjecture on X and Z, but does *not* resolve it for new values of r.

Proposition 20.1 (Hodge locus under blow-up). *Let $\pi : \widetilde{X} = \mathrm{Bl}_Z X \to X$ be a blow-up along a regular center of codimension c, satisfying (A)+(B). For each $r \geq 1$:*

1. $\mathrm{SFC}^r(\widetilde{X}) \cong \mathrm{SFC}^r(X) \oplus \bigoplus_{s=1}^{\min(r,c-1)} \mathrm{SFC}^{r-s}(Z)$.
2. *If Conjecture 18.1 holds for (X, r) and $(Z, r-s)$ for all $1 \leq s \leq \min(r, c-1)$, then it holds for $(\widetilde{X}, r)$.*

Proof. (1) The blow-up isomorphism is φ- and $\mathcal{N}^\bullet$-compatible (Proposition 5.1). Conditions (i)–(iv) decompose componentwise; the Tate twist shift $\mathcal{N}^s \to \mathcal{N}^{s+r}$ (Lemma 2.1) converts codimension-r on $\widetilde{X}$ to codimension-$(r-s)$ on Z for each summand. Terms with $s \geq \min(r, c)$ do not arise (Step 4 of the proof of Theorem 1.1). (2) Follows from (1) and Theorem 14.1. □

Corollary 20.2 (Stability under minimal resolutions). *Let S/O_K be a surface satisfying (A)+(B), $p > 3$, with Conjecture 18.1 holding for $r = 1$ on S (e.g. when S is an abelian surface, by Corollary 18.4). Then the conjecture holds for $r = 1$ on the A_n- and D_n-resolutions of Propositions 10.2 and 10.3.*

Proof. Each blow-up step has $c = 2$, contributing $\mathrm{SFC}^0(Z)$ for $Z = \operatorname{Spec} O_K$. By Remark 17.5, $\mathrm{SFC}^0(\operatorname{Spec} O_K) = \mathbb{Z}_p$, generated by $[\operatorname{Spec} O_K]$, so the conjecture holds trivially. Induction via Proposition 20.1(2) concludes. □

Example 20.3 (Cycle classes on A_n and D_n resolutions revisited). For $\pi : \widetilde{S} \to S$ an A_n-resolution, $\mathrm{SFC}^1(\widetilde{S}) \cong (\mathrm{NS}(S) \oplus \mathbb{Z}^n) \otimes \mathbb{Z}_p$ via $\mathrm{cl}_1^{\mathrm{Prism}}$. The n generators correspond to the exceptional classes $[E_i]$, and their cup products encode the A_n intersection form (Proposition 10.4), distinguishing A_n from D_n at the cycle level.

Revised status of the three problems from §16.

1. Prismatic Hodge conjecture. Proved for $r = 1$: abelian schemes (Theorem 18.3) and K3 surfaces with $p > 3$ (Theorem 18.5). Open for $r \geq 2$ (Remark 18.6); for surfaces the case $r = 2$ is related to the Tate conjecture for divisors. Stable under blow-up (Proposition 20.1) and under A_n/D_n resolutions (Corollary 20.2).

2. Torsion injectivity. Proved for $r = 1$ for abelian schemes and K3 surfaces (Theorem 19.3, Corollary 19.5). Conditional for $r \geq 2$ on hypothesis (C) (Proposition 19.6), which is open (Warning 19.7).

3. Prismatic motivic realization. The problem of a functor from Voevodsky's motives to $D^{\varphi,N}_{\mathrm{pris}}(A_{\mathrm{inf}})$ compatible with $\mathrm{cl}_r^{\mathrm{Prism}}$ remains open. Theorems 18.3 and 19.3 are independent of it.

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Appendix B

Expanded Proofs: Frobenius Overlap Verification and the Local-Global Obstruction

Purpose. This appendix supplies the two proofs that the referee (external review, May 2026) identified as insufficiently detailed in the main text:

(1) The verification that the local Frobenius lifts constructed in Lemma 6.2 agree on chart overlaps.

(2) The derivation that the local-global obstruction of Theorem 13.4 lands in $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}})$.

These appendix proofs *replace* the corresponding sketched arguments in the main text; all other statements of Lemma 6.2 and Theorem 13.4 remain unchanged.

B.1 Frobenius overlaps in the blow-up: proof of Lemma 6.2(A)

Setup. Let $\mathrm{Spec}(B) \subset X$ be an affine open in which the ideal of Z is generated by a regular sequence $t_1, \dots, t_c \in B$. Write $\varphi_B(t_i) = t_i^p + p\delta_i$ (with $\delta_i \in B$) for the given Frobenius lift on a smooth $W_2(k)$-lift $\widetilde{B}$ of B. The standard open cover of $\widetilde{E}_1 := \mathrm{Bl}_Z\mathrm{Spec}(B)$ consists of the charts

$$\widetilde{B}_l := B[s_1, \dots, \hat{s}_l, \dots, s_c]\big/(t_j - t_l s_j \mid j \neq l), \qquad l = 1, \dots, c,$$

where $s_j = t_j/t_l$ on the l-th chart. For $l = 1$ the Frobenius lift $\widetilde{\varphi}_1$ is defined by formula (4) of the main text:

$$\widetilde{\varphi}_1(b) = \varphi_B(b), \quad \widetilde{\varphi}_1(s_j) = s_j^p + p(\delta_j - \delta_1 s_j^p)t_1^{-p}, \quad j = 2, \dots, c. \tag{1}$$

For a general chart $l \geq 2$, the analogous formula is

$$\widetilde{\varphi}_l(b) = \varphi_B(b), \quad \widetilde{\varphi}_l(s_j) = s_j^p + p(\delta_j - \delta_l s_j^p)t_l^{-p}, \quad j \neq l. \tag{2}$$

Claim. $\widetilde{\varphi}_1$ and $\widetilde{\varphi}_l$ agree on the overlap $\widetilde{B}_1[s_l^{-1}] = \widetilde{B}_l[s_1^{-1}]$.

Proof of the claim. On the overlap $s_l \neq 0$, the two charts are identified by the change of coordinates

$$s_{1,l} := \frac{t_1}{t_l} = \frac{1}{s_l}, \quad s_{j,l} := \frac{t_j}{t_l} = \frac{s_j}{s_l} \text{ for } j \neq 1, l, \quad t_l = t_1 s_l, \tag{3}$$

where $s_{j,l}$ denotes the j-th coordinate on chart l. We verify that $\widetilde{\varphi}_1 = \widetilde{\varphi}_l$ under (3) in three steps, using $p^2 = 0$ throughout.

Step 1: agreement on $s_{1,l} = s_l^{-1}$. $\widetilde{\varphi}_1$ sends s_l to $\widetilde{\varphi}_1(s_l) = s_l^p + p(\delta_l - \delta_1 s_l^p)t_1^{-p}$. Write $\widetilde{\varphi}_1(s_l) = s_l^p(1 + p(\delta_l s_l^{-p} - \delta_1)t_1^{-p})$. Since $p^2 = 0$, inversion is $(1+px)^{-1} = 1 - px$, so

$$\widetilde{\varphi}_1(s_l)^{-1} = s_l^{-p}\big(1 - p(\delta_l s_l^{-p} - \delta_1)t_1^{-p}\big) = s_l^{-p} + p(\delta_1 - \delta_l s_l^{-p})s_l^{-p}t_1^{-p}.$$

Using $t_1^{-p}s_l^{-p} = (t_1 s_l)^{-p} = t_l^{-p}$ and $s_{1,l} = s_l^{-1}$:

$$\widetilde{\varphi}_1(s_l^{-1}) = s_{1,l}^p + p(\delta_1 - \delta_l s_{1,l}^p)t_l^{-p} = \widetilde{\varphi}_l(s_{1,l}).$$

Agreement on $s_{1,l}$ verified.

Step 2: agreement on $s_{j,l} = s_j/s_l$ for $j \neq 1, l$. Since $\widetilde{\varphi}_1$ is a ring homomorphism and $p^2 = 0$, for any units $u = s_l^p + p(\dots)$ and $a = s_j^p + p(\dots)$ one has $(a+pb)(u+pv)^{-1} = au^{-1} + p(bu^{-1} - avu^{-2}) \pmod{p^2}$. Set $a = \widetilde{\varphi}_1(s_j)$, $u = \widetilde{\varphi}_1(s_l)$:

$$\begin{aligned}\widetilde{\varphi}_1(s_j/s_l) &= \widetilde{\varphi}_1(s_j) \cdot \widetilde{\varphi}_1(s_l)^{-1} \\ &= \big(s_j^p + p(\delta_j - \delta_1 s_j^p)t_1^{-p}\big)\big(s_l^p + p(\delta_l - \delta_1 s_l^p)t_1^{-p}\big)^{-1}.\end{aligned}$$

The leading term is $(s_j/s_l)^p = s_{j,l}^p$. The p-correction: writing $A = s_j^p$, $B = (\delta_j - \delta_1 s_j^p)t_1^{-p}$, $U = s_l^p$, $V = (\delta_l - \delta_1 s_l^p)t_1^{-p}$,

$$\begin{aligned}BU - AV &= s_l^p(\delta_j - \delta_1 s_j^p)t_1^{-p} - s_j^p(\delta_l - \delta_1 s_l^p)t_1^{-p} \\ &= t_1^{-p}\big[\delta_j s_l^p - \delta_l s_j^p\big] \quad \text{(the } \delta_1\text{-terms cancel).}\end{aligned}$$

So the p-coefficient of $\widetilde{\varphi}_1(s_j/s_l)$ is

$$\frac{BU - AV}{U^2} = \frac{\delta_j s_l^p - \delta_l s_j^p}{s_l^{2p}} t_1^{-p} = \big(\delta_j s_l^{-p} - \delta_l s_j^p s_l^{-2p}\big)t_1^{-p} = \big(\delta_j - \delta_l (s_j/s_l)^p\big)(t_1 s_l)^{-p} = \big(\delta_j - \delta_l s_{j,l}^p\big)t_l^{-p}.$$

Therefore

$$\widetilde{\varphi}_1(s_{j,l}) = s_{j,l}^p + p(\delta_j - \delta_l s_{j,l}^p)t_l^{-p} = \widetilde{\varphi}_l(s_{j,l}).$$

Agreement on all $s_{j,l}$ verified.

Step 3: agreement on t_1. On chart l, we have $t_1 = t_l \cdot s_{1,l}$.

$$\begin{aligned}\widetilde{\varphi}_1(t_1) &= t_1^p + p\delta_1. \\ \widetilde{\varphi}_l(t_l \cdot s_{1,l}) &= \widetilde{\varphi}_l(t_l) \cdot \widetilde{\varphi}_l(s_{1,l}) \\ &= (t_l^p + p\delta_l)(s_{1,l}^p + p(\delta_1 - \delta_l s_{1,l}^p)t_l^{-p}).\end{aligned}$$

Using $p^2 = 0$ and $s_{1,l} = t_1/t_l$:

$$\begin{aligned}\widetilde{\varphi}_l(t_l \cdot s_{1,l}) &= t_l^p s_{1,l}^p + p t_l^p(\delta_1 - \delta_l s_{1,l}^p)t_l^{-p} + p\delta_l s_{1,l}^p \\ &= (t_l s_{1,l})^p + p\delta_1 - p\delta_l s_{1,l}^p + p\delta_l s_{1,l}^p \\ &= t_1^p + p\delta_1.\end{aligned}$$

Agreement on t_1 verified.

Since $\widetilde{B}_1[s_l^{-1}]$ is generated over B by $t_1, s_2, \dots, s_c$ (with $t_j = t_1 s_j$), Steps 1–3 cover all generators and the claim follows. By symmetry (exchanging the roles of 1 and l), all pairwise overlaps among the c charts are handled. □

Remark B.1. The key feature exploited is that $p^2 = 0$: the correction terms $p\delta_i$ are nilpotent, so inversion of Frobenius-translated coordinates is algebraically explicit. No deformation theory or cohomological gluing argument is required; the overlap condition reduces to an elementary identity in $\widetilde{B}_l[s_j^{-1}]$.

B.2 The local-global obstruction: proof of Theorem 13.4

Setup. Let $Z \in Z^r(X)$ be a codimension-r cycle such that each $Z \cap U_\alpha$ is cut out by a regular sequence on the affine open $U_\alpha \subset X$. For each α, the prismatic Koszul construction of [3, Sec. 9] yields a local cycle class $\mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\alpha}) \in H^{2r}_{\mathrm{Prism}}(U_\alpha)(r)$.

Step 1: the Čech 1-cocycle. On each double overlap $U_{\alpha\beta} := U_\alpha \cap U_\beta$, the restrictions $\mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\alpha})|_{U_{\alpha\beta}}$ and $\mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\beta})|_{U_{\alpha\beta}}$ are both local cycle classes for $Z|_{U_{\alpha\beta}}$, hence they agree after inverting p (since the rational prismatic cycle class is unique by the crystalline comparison and (P3)). Their difference

$$c_{\alpha\beta} := \mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\alpha})\big|_{U_{\alpha\beta}} - \mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\beta})\big|_{U_{\alpha\beta}} \;\in\; H^{2r}_{\mathrm{Prism}}(U_{\alpha\beta})(r)[p^\infty]$$

is a Čech 1-cocycle for the sheaf $\mathcal{H}^{2r} := H^{2r}_{\mathrm{Prism}}(-)(r)$ on X.

Step 2: from Čech to derived-category obstruction. Since $R\Gamma_{\mathrm{Prism}}(X)(r)$ is a perfect complex of A_{inf}-modules (P1), it admits a representative

$$F^\bullet : \quad \cdots \to F^{2r-1} \xrightarrow{d} F^{2r} \xrightarrow{d} F^{2r+1} \to \cdots$$

by bounded complexes of finitely generated *free* A_{inf}-modules. Set $Z^{2r} := \ker(d : F^{2r} \to F^{2r+1})$ and $B^{2r} := \mathrm{im}(d : F^{2r-1} \to F^{2r})$, so $H^{2r}(F^\bullet) = H^{2r}_{\mathrm{Prism}}(X)(r)$.

Each local class $\mathrm{cl}_r^{\mathrm{Prism}}(Z|_{U_\alpha})$ is represented by an element $f_\alpha \in Z^{2r}|_{U_\alpha}$. The cocycle $c_{\alpha\beta} = f_\alpha|_{U_{\alpha\beta}} - f_\beta|_{U_{\alpha\beta}}$ lies in $Z^{2r}|_{U_{\alpha\beta}}$ and maps to zero in $H^{2r}(F^\bullet)$, hence $c_{\alpha\beta} \in B^{2r}|_{U_{\alpha\beta}}$: there exists $e_{\alpha\beta} \in F^{2r-1}|_{U_{\alpha\beta}}$ with $d(e_{\alpha\beta}) = c_{\alpha\beta}$. The collection $(e_{\alpha\beta})$ defines a Čech 1-cochain for F^{2r-1}. Its coboundary $e_{\alpha\beta} - e_{\alpha\gamma} + e_{\beta\gamma}$ lies in $\ker d = Z^{2r-1}$, so it defines a Čech 1-cocycle $\eta \in \check{H}^1(\{U_\alpha\}, Z^{2r-1})$.

Step 3: identifying the obstruction group. Consider the short exact sequence of sheaves of A_{inf}-modules on X:

$$0 \;\to\; H^{2r-1}_{\mathrm{Prism}}(X)(r)_{\mathrm{sh}} \;\to\; F^{2r-1}/B^{2r-1} \;\xrightarrow{d}\; Z^{2r-1}/B^{2r-1} \cap Z^{2r-1} \;\to\; 0, \qquad (4)$$

where $(-)_{\mathrm{sh}}$ denotes the constant sheaf with value equal to the A_{inf}-module $H^{2r-1}_{\mathrm{Prism}}(X)(r)$. Since F^{2r-1} is a free A_{inf}-module sheaf, the middle term is flasque, hence $H^1(X, F^{2r-1}/B^{2r-1}) = 0$. The long exact cohomology sequence of (4) therefore gives an injection

$$\check{H}^1\big(\{U_\alpha\}, Z^{2r-1}\big) \;\hookrightarrow\; H^1\big(X,\, H^{2r-1}_{\mathrm{Prism}}(X)(r)_{\mathrm{sh}}\big).$$

The class η maps to a class $\mathrm{obs}(Z) \in H^1(X, H^{2r-1}_{\mathrm{Prism}}(X)(r)_{\mathrm{sh}})$.

Step 4: cohomology of a constant sheaf. Since X is an affine-locally trivial scheme and $H^{2r-1}_{\mathrm{Prism}}(X)(r)$ is a finitely generated A_{inf}-module (P1), the constant sheaf with value $M := H^{2r-1}_{\mathrm{Prism}}(X)(r)$ satisfies

$$H^1(X, M_{\mathrm{sh}}) \;=\; \mathrm{Ext}^1_{A_{\mathrm{inf}}}(M, A_{\mathrm{inf}})$$

by the universal coefficient exact sequence for the perfect complex $R\Gamma_{\mathrm{Prism}}(X)(r)$ over the local ring A_{inf} [1, Tag 0A5E]:

$$0 \to \mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}) \to H^1\big(X, H^{2r-1}_{\mathrm{Prism}}(X)(r)_{\mathrm{sh}}\big) \to_{A_{\mathrm{inf}}} (H^{2r}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}) \qquad (5)$$

(the left term is the Ext^1 of the previous cohomology group; see [1, Tag 0BYB] for the exact sequence for a 2-term truncation of a perfect complex). Thus $\mathrm{obs}(Z) \in \mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}})$ as claimed in the statement of Theorem 13.4.

Vanishing under (A)+(B). By Lemma 13.2, $H^{2r-1}_{\mathrm{Prism}}(X)(r)$ is a free A_{inf}-module of finite rank under (A)+(B). By Lemma 13.3, $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}) = 0$. Hence $\mathrm{obs}(Z) = 0$ and the local cycle classes glue to a global class in $H^{2r}_{\mathrm{Prism}}(X)(r)$.

Vanishing in the three special cases.
Case (i): Z rationally equivalent to zero. If $Z = \mathrm{div}(f)$ for a rational function f, then on each U_α the local cycle class equals the image of the divisor class of f, which is intrinsically defined (independent of the regular sequence chosen). Hence the local classes agree and $c_{\alpha\beta} = 0$ for all α, β, so $\mathrm{obs}(Z) = 0$.
Case (ii): $H^{2r-1}(X, \Omega^{r-1}_{X/O_K})$ is p-torsion-free. By the Hodge–Tate comparison (P5), $\mathrm{gr}^{r-1}_{\mathcal{N}} H^{2r-1}_{\mathrm{Prism}}(X)(r) \otimes_{A_{\mathrm{inf}},\theta} W(k) \cong H^r(X, \Omega^{r-1})$ is p-torsion-free. Combining with Lemma 13.1, which gives p-torsion-freeness of the full group $H^{2r-1}_{\mathrm{Prism}}(X)(r)$, one obtains $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}}) = 0$ by the same argument as Lemma 13.3 (torsion-free finitely generated A_{inf}-module is free by [1, Tags 0539, 058M, 00NX]).
Case (iii): $r = 1$. The prismatic first Chern class $c_1^{\mathrm{Prism}}(L)$ for a line bundle L is constructed globally in [3, Sec. 7]; no local-global ambiguity arises. ☐

Remark B.2 (Completeness status). The argument above fills the gap identified in §13 of the main text. The three pieces that were previously unclear are now explicit: (a) the Čech 1-cocycle in Step 1 is constructed without ambiguity once local lifts to F^{2r-1} are chosen; (b) the identification of the obstruction with an element of $\mathrm{Ext}^1_{A_{\mathrm{inf}}}(H^{2r-1}_{\mathrm{Prism}}(X)(r), A_{\mathrm{inf}})$ in Step 4 uses the standard universal coefficient sequence (5) from [1]; (c) vanishing under (A)+(B) follows immediately from Lemma 13.3. The three special-case vanishing conditions are now rigorously established.

References for Appendix B.

[3] B. Bhatt, P. Scholze, *Prisms and prismatic cohomology*, Ann. of Math. (2) 196 (2022), no. 3, pp. 1135–1275. (= *reference [3] of the main text.*)
[1] The Stacks Project Authors, *Stacks Project*, `https://stacks.math.columbia.edu`, 2024. Tag 0A5E: universal coefficient sequence for perfect complexes. Tag 0BYB: low-degree terms from Postnikov truncation. Tag 0539: flat ⇔ torsion-free over valuation rings. Tags 058M, 00NX: finitely generated projective over local ring is free. (= *reference [1] of the main text.*)